Math 104 Notes

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Contents

1 Introduction

1.1 Ordered Sets

Definition 1.1. Let S be a set. An order on S is a relation, denoted $<$ such that the following are true.

- If $x, y \in S$, then one and only one of $x < y$, $x = y$, and $y < x$ is true.
- If $x, y, z \in S$ and $x < y$, $y < z$, then $x < z$.

Notation. We may write $y > x$ instead of $x < y$. We write $x \leq y$ for $\neg(y < x)$.

Definition 1.2. An ordered set is a set S in which an order is defined.

Definition 1.3. Let S be an ordered set, and $E \subseteq S$. If there is a $\beta \in S$ such that $\forall x \in E, x \leq \beta$, we say that E is bounded above, and that β is an upper bound of E. We define sets being bounded below and lower bounds similarly.

Definition 1.4. Suppose S is an ordered set, $E \subseteq S$ is bounded above, and there is an $\alpha \in S$ that is an upper bound of E, such that if $\gamma < \alpha$, then γ is not an upper bound of E. Then, α is the least upper bound, or supremum of E. We write $\alpha = \sup E$. Similarly, we can have the greatest lower bound or infimum, for which we write $\inf E$.

Definition 1.5. We say that an ordered set S has the least-upper-bound property if, for all $E \subseteq S$ that is bounded above, $\sup E$ exists in S.

Theorem 1.6. *Suppose that S is an ordered set with the least-upper-bound property, $B \subseteq S$, B is nonempty, and B is bounded below. Let L be the set of all lower bounds of B. Then,*

$$\sup L = \alpha = \inf B$$

exists in S.

Proof. Because B is bounded below, we have that $L \neq \varnothing$. We can see that every $x \in B$ is an upper bound of L. This means that L is bounded above, and $\sup L := \alpha$ exists in S. If we let $\gamma < \alpha$, we must have that γ is not an upper bound of L, and therefore $\gamma \notin B$. This means that, for all $x \in B$, $\alpha \leq x$, and so $\alpha \in L$. Now, let us suppose that $\alpha < \beta$. Then, $\beta \notin L$, because α is an upper bound of L. Therefore, α is a lower bound of B, but if $\beta > \alpha$, then β is not, meaning that $\alpha = \inf B$. ■

1.2 Ordered Fields

Definition 1.7. An ordered field is a field F which is also an ordered set, such that

- For $x, y, z \in F$, if $y < z$, then $x + y < x + z$
- For $x, y \in F$ with $x, y > 0$, $xy > 0$.

If $x > 0$, we say it is positive, and if $x < 0$ we say that it is negative.

Proposition 1.8. *The following are true.*

(1) If $x > 0$, $-x < 0$ and vice versa.

(2) If $x > 0$ and $y < z$, then $xy < xz$.

(3) If $x < 0$ and $y < z$, then $xy > xz$.

(4) If $x \neq 0$, then $x^2 > 0$.

(5) If $0 < x < y$, then $0 < y^{-1} < x^{-1}$.

Proof.

(1) Subtract x from both sides.

(2) This is the same as saying $z - y > 0$. Thus, $xz - xy > 0$, or $xy < xz$.

(3) We have that $(-x)y < (-x)z$. Adding $xy + xz$ to both sides gives $xy > xz$.

(4) If $x > 0$, then (2) implies that $x^2 > 0$. If $x < 0$, then $-x > 0$, so $(-x)^2 = x^2 > 0$.

(5) Note that $1 > 0$ is a consequence of (4). Now, suppose that $y^{-1} < 0$. Multiplying both sides by y gives $1 < 0$, a contradiction. Thus, $0 < y^{-1}$, and similarly for x^{-1}. Multiplying by $x^{-1}y^{-1}$ gives that $0 < y^{-1} < x^{-1}$.

■

1.3 The Real Numbers

Theorem 1.9. *There exists an ordered field $\mathbb{R}$ with the least-upper-bound property such that $\mathbb{Q}$ is a subfield of $\mathbb{R}$. The members of $\mathbb{R}$ are called the Real Numbers.*

Proof. In order to prove this, we must define a cut in $\mathbb{Q}$. We define a cut in $\mathbb{Q}$ to be a pair of subsets of $\mathbb{Q}$, A and B, such that

- $A \cup B = \mathbb{Q}$, $A \neq \varnothing$, $B \neq \varnothing$, $A \cap B = \varnothing$
- If $a \in A$ and $b \in B$ then $a < b$
- A contains no largest element.

We will write $A|B$ for this cut. We define a real number to be a cut in $\mathbb{Q}$. We shall first define an order on this. Given cuts $A|B$ and $C|D$, we say that $A|B \leq C|D$ if $A \subseteq C$, and $A|B < C|D$ if $A \subset C$. We define addition as $A|B + C|D = E|F$ where

$$E = \{r \in \mathbb{Q} : r = a + c \text{ for some } a \in A, c \in C\}$$

$$F = \mathbb{Q} \setminus E.$$

It is clear that this is commutative, associative and closed. We can also see that

$$0^* = \{r \in \mathbb{Q} : r < 0\}|\{r \in \mathbb{Q} : r \geq 0\}$$

is the additive identity, and that the additive inverse of $A|B$ is $C|D$ where

$$C = \{r \in \mathbb{Q} : r = -b \text{ for some } b \in B \text{ that is not the smallest element}\}$$

$$D = \mathbb{Q} \setminus C.$$

If a cut $A|B > 0^*$ we say it is positive, and if $A|B < 0^*$, we say it is negative. We now define multiplication between two positive cuts $A|B$ and $C|D$ to be $E|F$ where

$$E = \{r \in \mathbb{Q} : r \leq 0 \text{ or } \exists a \in A, c \in C \text{ such that } a > 0, c > 0, r = ac\}$$

$$F = \mathbb{Q} \setminus E.$$

We can also see that the multiplicative identity is

$$1^* = \{r \in \mathbb{Q} : r < 1\}|\{r \in \mathbb{Q} : r \geq 1\},$$

and that the multiplicative inverse of $A|B$ is $C|D$ where

$$C = \{r \in \mathbb{Q} : r \leq 0 \text{ or } r = b^{-1} \text{ for some } b \in B \text{ that is not the smallest element}\}$$

$$D = \mathbb{Q} \setminus C.$$

We can also see associativity, commutativity, and closure. We now extend the definition of multiplication to be

$$A|B \cdot C|D = \begin{cases} (A|B) \cdot (C|D) & A|B > 0^*, C|D > 0^* \\ -((A|B) \cdot (-C|D)) & A|B > 0^*, C|D < 0^* \\ -((-A|B) \cdot (C|D)) & A|B < 0^*, C|D > 0^* \\ (-A|B) \cdot (-C|D) & A|B < 0^*, C|D < 0^* \\ 0^* & A|B = 0^* \text{ or } C|D = 0^* \end{cases}.$$

We will not prove that this is a field because it is simply too tedious to check every case. Assuming that it does form a field (which it does), we can see that it is in fact an ordered field. Notice that we may also associate each $r \in \mathbb{Q}$ with

$$r^* = \{q \in \mathbb{Q} : q < r\}|\{q \in \mathbb{Q} : q \geq r\},$$

and that these elements form a subfield that is isomorphic to $\mathbb{Q}$. Finally, we show that this has the least upper bound property. We will let $\mathcal{C} \subset \mathbb{R}$ be nonempty and bounded above by $X|Y$. We define

$$C = \{r \in \mathbb{Q} : \text{for some } A|B \in \mathcal{C}, r \in A\}$$

$$D = \mathbb{Q} \setminus C.$$

We can see that $C|D$ is an upper bound of $\mathcal{C}$. Let us suppose that $C'|D'$ is an upper bound of $\mathcal{C}$. By definition, we have that $A \subseteq C'$ for all $A|B \in \mathcal{C}$. Therefore, $C \subseteq C'$, meaning that $C|D \leq C'|D'$. Thus, among all the upper bounds of $\mathcal{C}$, $C|D$ is the least. ■

Theorem 1.10 (The Archimedean Property)**.** *If $x, y \in \mathbb{R}$ and $x > 0$, then there is a positive integer n such that $nx > y$.*

Proof. Let $A = \{nx : n \in \mathbb{N}\}$. If this were false, then y would be an upper bound of A. We let $\alpha = \sup A$. Because $x > 0$, we have that $\alpha - x < \alpha$, meaning that $\alpha - x$ is not an upper bound of A. Thus, there is some m such that $\alpha - x < mx$, meaning that $\alpha < (m+1)x$, a contradiction. ■

Theorem 1.11. *If $x, y \in \mathbb{R}$ and $x < y$, then there is a $p \in \mathbb{Q}$ such that $x < p < y$.*

Proof. We have that $y - x > 0$, meaning that there is some positive integer n such that

$$n(y - x) > 1.$$

We also have that there are integers m_1 and m_2 such that $m_1 > nx$ and $m_2 > -nx$. Therefore, we have that

$$-m_2 < nx < m_1.$$

Changing the first inequality to allow equality, we may increase (decrease) our lower (upper) bound so that we have the largest (smallest) integer lower (upper) bound, giving us $m_2' \leq nx < m_1'$. We claim that $m_1' = m_2' + 1$. To see this, note that we must have

$$m_2' < m_2' + 1 \leq m_1',$$

and that nx can only appear between m_2' and $m_2' + 1$ without contradicting m_2''s maximality. Thus, by the minimality of m_1', $m_1' = m_2' + 1$. Writing m for m_1' gives

$$m - 1 \leq nx < m.$$

Combining this with our original inequality gives

$$nx < m \leq 1 + nx < ny,$$

meaning that $x < \frac{m}{n} < y$. ■

Theorem 1.12. *For every real $x > 0$ and every integer $n > 0$, there exists a unique positive real y such that $y^n = x$.*

Proof. Uniqueness is clear because, if $0 < y_1 < y_2$ then $y_1^n < y_2^n$. Now, let $E = \{t \in \mathbb{R}_{>0} : t^n < x\}$. Note that $t_0 = \frac{x}{x+1}$ has the property that $0 \leq t_0 < 1$, meaning that $t_0^n \leq t_0 < x$. Therefore, E is nonempty. Note also that $t_1 = 1 + x$ has the property that $t_1^n \geq t_1 > x$, meaning that $1 + x$ is an upper bound of E. Therefore, $y = \sup E$ exists. We claim that this is $\sqrt[n]{x}$. To show this, we will show that $y^n > x$ and $y^n < x$ yield contradictions.

When $0 < a < b$, we can use the fact that $b^n - a^n = (b - a)(b^{n-1} + b^{n-2}a + \cdots + a^{n-1})$ to get that

$$b^n - a^n < (b - a)nb^{n-1}.$$

Let us suppose that $y^n < x$. We will chose h such that

$$0 < h < \min\left(1, \frac{x - y^n}{n(y + 1)^{n-1}}\right).$$

Putting $a = y$ and $b = y + h$ into our inequality gives

$$(y + h)^n - y^n < hn(y + h)^{n-1} < hn(y + 1)^{n-1} < x - y^n,$$

meaning that $(y + h)^n < x$, and is thus in E. However, we chose $y = \sup E$, and $y + h > y$, so this is a contradiction.

Let us assume that $y^n > x$. We will let

$$k = \frac{y^n - x}{ny^{n-1}},$$

which has the property $0 < k < y$. Let $t \geq y - k$. Then,

$$y^n - t^n \leq y^n - (y + k)^n < kny^{n-1} = y^n - x.$$

This means that $x < t^n$. Thus, $y - k$ is an upper bound of E, contradicting the fact that $y = \sup E$. Therefore, $y^n = x$. ■

2 Basic Topology

2.1 Metric Spaces

Definition 2.1. Let M be a set. A metric on M is a function $d : M \times M \to \mathbb{R}_{\geq 0}$ such that

- $\forall x, y \in M, d(x, y) = 0 \iff x = y$
- $\forall x, y \in M, d(x, y) = d(y, x)$
- $\forall x, y, z \in M, d(x, z) \leq d(x, y) + d(y, z)$.

In this case, (M, d) is called a metric space.

Example 2.2. In $\mathbb{R}^k$, $d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}|$ is a metric.

Example 2.3. Let S be a set. The discrete metric, defined by

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

is a metric on S.

2.2 Open and Closed Sets

Definition 2.4. Suppose that (M, d) is a metric space, $x \in M$, and $r > 0$. The open ball of radius r centered at x is the set

$$B_r(x) = \{y \in M : d(x, y) < r\}.$$

Example 2.5. Let (M, d) be the discrete metric space. Then,

$$B_r(x) = \begin{cases} M & r > 1 \\ x & r \leq 1 \end{cases}.$$

Definition 2.6. Let (M, d) be a metric space, and $E \subseteq M$. A point $x \in E$ is called an interior point of E if there is some $r > 0$ such that $B_r(x) \subseteq E$.

Definition 2.7. Let (M, d) be a metric space. A set $U \subseteq M$ is said to be open if, for all $u \in U$, there exists $r > 0$ such that $B_r(u) \subseteq U$. In other words, all $u \in U$ are interior points.

Example 2.8. In the discrete metric space, every set is open.

Proposition 2.9. *Open balls are open.*

Proof. Let us fix $y \in B_r(x)$. Let $s = r - d(x, y)$. If $z \in B_s(y)$, then we have that $d(y, z) < s = r - d(x, y)$, meaning that $d(x, z) \leq d(x, y) + d(y, z) < r$. Thus, $z \in B_r(x)$, so $B_s(y) \subseteq B_r(x)$. ■

Definition 2.10. Suppose that (M, d) is a metric space and that $E \subseteq M$. A limit point of E is a point $x \in M$ such that, for all $r > 0$, $B_r(x) \setminus \{x\}$ contains a point of E. The set of all limit points of E is denoted E'.

Example 2.11. If $\mathbb{R}$, $(0, 1)' = [0, 1]$.

Example 2.12. In $\mathbb{R}$, $E' = \{0\}$, where $E = \{\frac{1}{n} : n \in \mathbb{N}\}$.

Definition 2.13. A set $E \subseteq M$ is said to be closed if $E' \subseteq E$.

Proposition 2.14. *Both M and $\varnothing$ are always open and closed.*

Proof. We will show this for M first. By definition, the limit points of M are in M, so M is closed. Similarly, the open balls only contain points of M, so M is also open. Now, we show this for $\varnothing$. Limit points must have balls around them that contain points of $\varnothing$, but there are no points in $\varnothing$ for these to contain, so it has no limit points. Thus, $\varnothing$ contains all of its limit points and is therefore closed. Now, in order for it not to be open, we must have a point $x \in \varnothing$ such that $B_r(x) \not\subseteq \varnothing$ for all r. However, there are no points in $\varnothing$, so this is impossible and $\varnothing$ is open. ■

Theorem 2.15. *Let (M, d) be a metric with $E \subseteq M$. Then, E is open if and only if E^c is closed.*

Proof. Let us assume that E^c is closed. Let $x \in E$. Therefore, x is not a limit point of E^c, so there is some $r > 0$ such that $B_r(x) \cap E^c = \varnothing$, meaning that $B_r(x) \subseteq E$, so that E is open. Now, let us suppose that E is open, and chose x to be a limit point of E^c. Then every ball centered at x contains a point of E^c, so x is not an interior point of E. Because E is open, we must therefore have that $x \in E^c$, meaning that E^c is closed. ■

Theorem 2.16. *Arbitrary unions and finite intersections of open sets are open. Arbitrary intersections and finite unions of closed sets are closed.*

Proof. Suppose that $(U_\alpha)_\alpha$ is a collection of open sets in the metric space (M, d). Let $x \in \bigcup_\alpha U_\alpha$. Then there is some β such that $x \in U_\beta$. This is open, so there is some $r > 0$ such that $B_r(x) \subseteq U_\beta \subseteq \bigcup_\alpha U_\alpha$, meaning that the union is open. Now, suppose that $U_1, U_2, \ldots, U_n$ are open, and let $x \in U_1 \cap U_2 \cap \cdots \cap U_n$. Then, for each U_i, there is some $r_i > 0$ such that $B_{r_i}(x) \subseteq U_i$. Choose $r = \min(r_1, r_2, \ldots, r_n)$. Because $r \leq r_i$ for each i, we have that $B_r(x) \subseteq B_{r_i}(x)$ for each i, so this is contained in every U_i. Therefore, $B_r(x) \subseteq \bigcap_{i=1}^n U_i$. Using the fact that

$$\bigcup_\alpha U_\alpha^c = \left(\bigcap_\alpha U_\alpha\right)^c$$

$$\bigcap_\alpha U_\alpha^c = \left(\bigcup_\alpha U_\alpha\right)^c,$$

we can see the analog of this for closed sets. ■

Example 2.17.

$$\bigcap_{t \in \mathbb{R}_{>1}} (-t, t) = [-1, 1].$$

Example 2.18.

$$\bigcap_{n=1}^{\infty} (-1 - \frac{1}{n}, 1 + \frac{1}{n}) = [-1.1].$$

Example 2.19. The set $\{t\} \subset \mathbb{R}$ is closed. Thus, if $E \subseteq \mathbb{R}$, then

$$E = \bigcup_{t \in E} \{t\},$$

so any set can be written as the union of closed sets.

Definition 2.20. Suppose that (M, d) is a metric space with $E \subseteq M$. The closure of E is the set $\overline{E} = E \cup E'$. The interior of E is the set

$$\overset{\circ}{E} = \{x \in E : \exists r > 0, B_r(x) \subseteq E\}.$$

E is said to be dense in M if $\overline{E} = M$.

Proposition 2.21. *$\mathbb{Q}$ is dense in $\mathbb{R}$.*

Proof. Suppose that $x \in \mathbb{R}$ and that $r > 0$. Then, there is some $q \in \mathbb{Q}$ with $x < q < x + r$, meaning that $q \in B_r(x) \setminus \{x\}$. Because this is true for any r, we have that $x \in \mathbb{Q}'$. Thus, $\mathbb{R} \subseteq \mathbb{Q}'$. However, $\mathbb{R}$ is the universal set so $\mathbb{Q}' \subseteq \mathbb{R}$, meaning that $\mathbb{Q}' = \mathbb{R}$. Therefore, $\mathbb{Q} \cup \mathbb{Q}' = \mathbb{R}$, so $\mathbb{Q}$ is dense in $\mathbb{R}$. ■

Theorem 2.22. *Let (M, d) be a metric space with $E \subseteq M$. Then the following are true.*

(1) $\overline{E}$ is closed.

(2) If E is closed, $E = \overline{E}$.

(3) If $F \subseteq M$ is closed and $E \subseteq F$, then $\overline{E} \subseteq F$.

Proof.

(1) Let us suppose that $x \in (\overline{E})'$. We wish to show that either $x \in E$ or $x \in E'$. We will let $\varepsilon > 0$. By the definition of a limit point, there is some $y \in B_\varepsilon(x) \setminus \{x\} \cap \overline{E}$. If $y \notin E$, then $y \in E'$, meaning that there is a point $z \in B_r(y) \setminus \{y\} \cap E$ where $r = \varepsilon - d(x, y)$. By the triangle inequality, we therefore have that $z \in B_\varepsilon(x)$, so $x \in E$ or $x \in E'$.

(2) If E is closed, then $E' \subseteq E$, meaning that $E = E \cup E' = \overline{E}$.

(3) Notice that, if $x \in E'$, then $x \in F'$. Because F is closed, we therefore have that $E' \subseteq F' \subseteq F$. Thus, $\overline{E} = E \cup E' \subseteq F$.

■

2.3 Boundedness and Compactness

Definition 2.23. Let (M, d) be a metric space with $E \subseteq M$. We say that E is *bounded* if there is some $r > 0$ such that, for all $x, y \in E$, $d(x, y) < r$.

Proposition 2.24. *Let (M, d) be a nonempty metric space with $E \subseteq M$. Then, the following are equivalent.*

(1) E is bounded.

(2) $\exists x \in M, \exists r > 0, E \subseteq B_r(x)$.

(3) $\forall x \in M, \exists r > 0, E \subseteq B_r(x)$.

Proof. We first show that (1) $\Rightarrow$ (3). Let $x \in M$. If $E = \varnothing$, then this is trivially satisfied, so we choose $E \neq \varnothing$ with $y \in E$. We will let $r_0 > 0$ be such that $d(y, z) < r_0$ for all $z \in E$. If we let $r = r_0 + d(x, y)$, then we have that $d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + r_0 = r$. From here, seeing that (3) $\Rightarrow$ (2) is trivial, so we only show that (2) $\Rightarrow$ (1). Let us suppose that we have $x \in M$ and $r > 0$ with $E \subseteq B_r(x)$. Let $y, z \in E$. Then, we have that $d(y, z) \leq d(x, y) + d(x, z) < r + r = 2r$, meaning that E is bounded. ■

Proposition 2.25. *Let $E \subseteq \mathbb{R}$. Then, E is bounded in the ordered set sense (bounded above and bounded below) if and only if it is bounded in the metric sense.*

Proof. Let us suppose that E is bounded in the set sense. Then, $E \subseteq B_r(\sup E)$ where $R = \sup E - \inf E + 1$. Now, let us suppose that E is bounded in the metric space sense. If $E \subseteq B_r(x)$, we have that E is bounded above by $x + r$, and bounded below by $x - r$. ■

Theorem 2.26. *Let $E \subseteq \mathbb{R}$ be bounded above and non-empty. Then,* $\sup E \in \overline{E}$.

Proof. If $\sup E \in E$, then this is clear. Let us suppose that $\sup E \notin E$, and that $\sup E$ is not a limit point. Then, there is some $r > 0$ such that $B_r(\sup E)$ does not contain any points of E. Consider $\sup E - \frac{r}{2}$. This is in $B_r(\sup E)$, and is therefore not in E. It must also be an upper bound of E (because otherwise there would be a point of E in $B_r(\sup E)$). However, this contradicts the minimality of $\sup E$, so $\sup E$ must be a limit point of E. ■

Definition 2.27. Let (M, d) be a metric space and $E \subseteq M$. An open cover of E is a set, U, of open subsets of M such that $E \subseteq \bigcup_{x \in U} x$.

Example 2.28. Let (M, d) be a metric space and $E \subseteq M$. The set $\{M\}$ is always an open cover of E.

Example 2.29. In $\mathbb{R}^2$, then an open cover of the set

$$S = \{(x, y) : 0 \leq x, y \leq 1\}$$

is the set

$$U = \{B_1(0,0), B_1(0,1), B_1(1,0), B_1(1,1)\}.$$

Definition 2.30. Let (M, d) be a metric space, and $E \subseteq M$. Then, E is said to be *compact* if every open cover of E has a finite subcover. In other words, if U is an open cover of E, then there is some $V \subseteq U$ that has finite cardinality and is also an open cover of E.

Proposition 2.31. *All finite sets are compact.*

Proof. Let $E = \{x_1, x_2, \ldots, x_n\} \subseteq M$ and let U be an open cover of E. For each x_i, there is some $G_i \in U$ with $x_i \in G_i$. Therefore,

$$E \subseteq \bigcup_{i=1}^{n} G_i,$$

so $\bigcup_{i=1}^{n} G_i$ is a finite subcover of E. ■

Example 2.32. $\left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup \{0\} \subseteq \mathbb{R}$ is compact. To see this, suppose that U is an open cover of the set (say, E). There must therefore be some open set $G \in U$ containing 0. However, this contains all but finitely many elements of E. Thus, if we have that $1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}$ are not in G, and if we define G_i to be the set containing i^{-1}, then $\{G, G_1, G_2, \ldots, G_n\}$ is a finite subcovering of E.

Example 2.33. $E = \{\frac{1}{n} : N \in \mathbb{N}\} \subseteq \mathbb{R}$ is not compact. To see this, let

$$U = \left\{ \left(\frac{\frac{1}{n+1} + \frac{1}{n}}{2}, \frac{\frac{1}{n} + \frac{1}{n-1}}{2} \right) : n \in \mathbb{N}, n \geq 2 \right\} \cup (0.9, 1.1).$$

This is a open covering of E, where each element contains only one element of E. Thus, there is no finite subcovering of U.

Proposition 2.34. *Let (M, d) be a metric space, and let $K \subseteq M$ be compact. Then, K is closed.*

Proof. Suppose that we have $x \notin K$. We let

$$U = \{\{y \in M : d(y, x) > r\} : r > 0\}.$$

First, note that $G_r = \{y \in M : d(y, x) > r\}$ is open. Now let $y \in K$. Because $x \notin K$, $d(y, x) > 0$. Then, $y \in G_{d(y,x)/2}$. This means that U is an open cover of K. Then, there are finitely many $r_1, r_2, \ldots, r_n$ such that $K \subseteq G_{r_1} \cup \cdots \cup G_{r_n}$. Note that, if $r = \min(r_1, \ldots, r_n)$, then $G_{r_1} \cup \cdots \cup G_{r_n} = G_r$. We have that $K \subseteq G_r$. Then, $B_r(x) \cap K \subseteq B_r(x) \cap G_r = \varnothing$. Therefore, $x \notin \overline{K}$. Thus, $\overline{K} \subseteq K \subseteq \overline{K}$, meaning that $K = \overline{K}$, and K is closed. ■

Proposition 2.35. *Compact sets are bounded.*

Proof. Let K be compact with $x \notin K$. Let

$$U = \{B_r(x) : r \in \mathbb{R}_{>0}\}.$$

This must be a open cover of K, because if $y \in K$ with $d(x, y) = r$, then $y \in B_{r+1}(x) \in U$. Thus, there must be some finite $r_1, r_2, \ldots, r_n$ with $K \subseteq B_{r_1}(x) \cup B_{r_2}(x) \cup \cdots \cup B_{r_n}(x)$. Choose $r = \max(r_1, r_2, \ldots, r_n)$. Then, $K \subseteq B_{r_1}(x) \cup B_{r_2}(x) \cup \cdots \cup B_{r_n}(x) = B_r(x)$. This means that, if $y \in K$, then $d(x, y) < r$. If we also have $z \in K$, then $d(y, z) \leq d(x, y) + d(x, z) < 2r$, so K is bounded. ∎

Proposition 2.36. *Closed subsets of compact sets are compact.*

Proof. Suppose that K is compact and that $F \subseteq K$ is closed. Let U be an open cover of F. Since F is closed, F^c is open. Let $V = U \cup \{F^c\}$. Note that V is an open cover of K (in fact, of M) since if $x \in K$, either $x \in F$ so covered by some element of U or $x \in F^c \in V$. Then, there are finitely many $G_1, \ldots, G_n \in V$ which cover K. Then $\{G_1, \ldots, G_n\} \setminus \{F^c\} \subseteq U$ is a finite subcover of F, for if $y \in F \subseteq K$, $y \in G_i$ for some i, and that $G_i \neq F^c$ because $y \in F$. ∎

Proposition 2.37. *Suppose that K is compact and $S \subseteq K$ is infinite. Then, S has a limit point in K.*

Proof. Suppose that S is a subset of K with no limit points in K. For each $x \in K$, let $r_x > 0$ be so that $B_{r_x}(x) \setminus \{x\} \cap S = \varnothing$. Now, let

$$U = \{B_{r_x}(x) : x \in K\}.$$

This is an open cover of K. Then, there are $x_1, x_2, \ldots, x_n \in K$ so that $S \subseteq K \subseteq B_{r_{x_1}}(x_1) \cup \cdots \cup B_{r_{x_n}}(x_n)$. Then, $S \subseteq \{x_1, x_2, \ldots, x_n\}$ since no $B_{r_{x_i}}(x_i)$ contains any point of S, except possibly x_i. Then, S is finite. ∎

Proposition 2.38. *If $\{K_\alpha\}$ is a collection of compact subsets of a metric space (M, d) such that the intersection of every finite subcollection of $\{K_\alpha\}$ is nonempty, then $\bigcap K_\alpha$ is nonempty.*

Proof. Let $K_1 \in \{K_\alpha\}$. We write $G_\alpha = K_\alpha^c$. Let us suppose that no point of K_1 belongs to every K_α. Then, $\{G_\alpha\}$ forms an open cover of K_1. Because K_1 is compact, there are finitely many $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that $K_1 \subseteq G_{\alpha_1} \cup G_{\alpha_2} \cup \cdots \cup G_{\alpha_n}$. However, this then means that

$$K_1 \cap K_{\alpha_1} \cap K_{\alpha_2} \cap \cdots \cap K_{\alpha_n} = \varnothing,$$

a contradiction. ∎

Proposition 2.39. *If $\{I_n\}$ is a sequence of (closed) intervals in $\mathbb{R}$, such that $I_n \supseteq I_{n+1}$ for $n \in \mathbb{N}$. Then, $\bigcap_{n=1}^{\infty} I_n \neq \varnothing$.*

Proof. Let $I_n = [a_n, b_n]$, and let $E = \{a_i\}$. This is nonempty and bounded above by b_1. Let $x = \sup E$. If $m, n \in \mathbb{Z}^+$, then

$$a_m \leq a_{m+n} \leq b_{m+n} \leq b_m,$$

meaning that $x \leq b_m$ for each m. Because $a_m \leq x$, we have that $x \in I_m$ for $m \in \mathbb{N}$. ■

Proposition 2.40. *Intervals $[a, b] \subset \mathbb{R}$ are compact.*

Proof. Let $\{U_i\}_{i \in I}$ be an open cover of $[a, b]$. This covers $[a, x]$ for every $x \in [a, b]$. Let S be the set of $x \in [a, b]$ such that $[a, x]$ is covered by finitely many U_i's. It is easy to see that $a \in S$, so S is a nonempty subset of $\mathbb{R}$ bounded above by b. We define

$$x_0 := \sup S \in [a, b].$$

Now, suppose that $x_0 \neq b$ (i.e. $x_0 < b$). Note that $x_0 > a$, as there exists $\varepsilon > 0$ and j such that $[a, a + \varepsilon] \subseteq U_j$ (by the definition of an open set). Take i_0 such that $x \in U_{i_0}$, and take $\varepsilon > 0$ such that $a \leq x_0 - \varepsilon < x_0 < x_0 + \varepsilon \leq b$ and $[x_0 - \varepsilon, x_0 + \varepsilon] \subseteq U_{i_0}$. Because $x_0 - \varepsilon$ is not an upper bound of S, there is some $x_0 - \varepsilon \leq x_1 \leq x_0$ such that $x_1 \in S$. Thus,

$$[a, x_1] \subseteq \bigcup_{j=1}^{n} U_{i_j} \Rightarrow [a, x_0 + \varepsilon] \subseteq \bigcup_{j=1}^{n} U_{i_j} \cup U_{i_0},$$

meaning $x_0 + \varepsilon \in S$, which is a contradiction. Thus, $\sup S = b$. We can also see that $b \in S$. We have that $[b - \varepsilon, b] \subseteq U_{i_0}$ for some i_0 and some $\varepsilon > 0$. There is thus some $x_1 \in [b - \varepsilon, b]$ such that $x_1 \in S$. Because $[a, x_1]$ has a finite subcover and $[b - \varepsilon, b] \subseteq U_{i_0}$, there is a finite subcover of $[a, b] = [a, x_1] \cup [b - \varepsilon, b]$. ■

Definition 2.41. A k-cell is a subset of $\mathbb{R}^k$ of the form

$$\begin{aligned}
&[a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_k, b_k] \\
&= \left\{ x \in \mathbb{R}^k : \substack{a_1 \leq x_1 \leq b_1 \\ a_2 \leq x_2 \leq b_2 \\ \vdots \\ a_k \leq x_k \leq b_k} \right\}.
\end{aligned}$$

Proposition 2.42. *Let $\{I_n\}$ be a sequence of k-cells such that $I_n \supset I_{n+1}$ for $n \in \mathbb{N}$. Then, $\bigcap_{n=1}^{\infty} I_n \neq \varnothing$.*

Proof. Let

$$I_n = \left\{ x \in \mathbb{R}^k : \substack{a_{n,1} \leq x_1 \leq b_{n,1} \\ a_{n,2} \leq x_2 \leq b_{n,2} \\ \vdots \\ a_{n,k} \leq x_k \leq b_{n,k}} \right\}.$$

We also let $I_{n,j} = [a_{n,j}, b_{n,j}]$. For each j, we also have $I_{n,j} \supset I_{n,j+1}$, so there is some $x_j \in \bigcap_{n=1}^{\infty} I_{n,j}$. Let $x = (x_1, x_2, \ldots, x_k)$. Then, $x \in I_n$ for $n \in \mathbb{N}$. ■

Proposition 2.43. *Every k-cell is compact.*

Proof. Let

$$I = \left\{ x \in \mathbb{R}^k : \begin{smallmatrix} a_1 \le x_1 \le b_1 \\ a_2 \le x_2 \le b_2 \\ \vdots \\ a_k \le x_k \le b_k \end{smallmatrix} \right\}.$$

Define also

$$\delta = \sqrt{\sum_{j=1}^{k} (b_j - a_j)^2}.$$

Then, if $x, y \in I$, $|x - y| \le \delta$. Now, suppose that there is some open cover $\{G_\alpha\}$ of I with no finite subcover. Let $c_j = \frac{a_j + b_j}{2}$. The intervals $[a_j, c_j]$ and $[c_j, b_j]$ thus determine 2^k k-cells whose union is I. At least one of these cannot be covered by a finite subcollection of $\{G_\alpha\}$. Call this I_1. We similarly divide I_1 to get a sequence $\{I_n\}$ such that

(a) $I \supset I_1 \supset I_2 \supset \cdots$,

(b) I_n is not covered by any finite subcollection of $\{G_\alpha\}$,

(c) If $x, y \in I_n$, then $|x - y| \le 2^{-n}\delta$.

Because of (a), there is some point x which lies in every I_n. For some α, $x \in G_\alpha$. Because G_α is open, there is some $r > 0$ such that $|y - x| < r$ means that $y \in G_\alpha$. However, if n is large enough that $2^{-n}\delta < r$, then (c) implies that $I_n \subset G_\alpha$. However, this contradicts (b). ■

Theorem 2.44 (The Heine-Borel Theorem). *If $K \subseteq \mathbb{R}^n$, the following are equivalent.*

(1) K is closed and bounded.

(2) K is compact.

(3) Every infinite subset of K has a limit point in K.

Proof. We first show that (1) $\Rightarrow$ (2). Suppose that $K \subseteq \mathbb{R}^n$ is closed and bounded. Let $r > 0$ be large enough that $K \subseteq B_r(0) \subseteq \{x \in \mathbb{R}^n : -r \le x_i \le r\}$. However,

$$\{x \in \mathbb{R}^n : -r \le x_i \le r\} = \underbrace{[-r, r] \times [-r, r] \times \cdots \times [-r, r]}_{n \text{ times}},$$

which is compact. Thus K is a closed subset of a compact set, and is therefore compact. We have already shown that (2) $\Rightarrow$ (3), so we only show that (3) $\Rightarrow$ (1). Suppose that K is not closed. Then, there is some $y \in K' \setminus K$. For each $n \in \mathbb{N}$, let $a_n \in B_{1/n}(y) \cap K$, and let $A = \{a_n : n \in \mathbb{N}\} \subseteq K$. Note that A is infinite. This is because, if A were finite, we could choose $r = \min(d(a, y) : a \in A) > 0$, but if $\frac{1}{N} < r$, then $d(a_N, y) < r$, which is a contradiction. Now suppose that $x \in K$. Consider $B_{d(x,y)/2}(x)$. If $N > \frac{2}{d(x,y)}$, then $a_n \in B_{d(x,y)/2}(y)$, meaning $d(a_N, x) > \frac{d(x,y)}{2}$. Thus, $B_{d(x,y)/2}(x)$ contains only finitely many points of A, so x is not a limit point of A. Therefore, $A' \cap K = \varnothing$. However, this contradicts (3), so K must be closed. Now, suppose that K is unbounded, and let $x_0 \in K$. For each $n \in \mathbb{N}$, choose $b_n \in K \setminus B_n(x_0)$. We define $\mathscr{B} = \{b_n : n \in \mathbb{N}\} \subseteq K$. Now, any finite set is contained in $B_n(x_0)$ for large enough n, but $\mathscr{B}$ contains points outside any such ball, so it must be infinite. Now, let $x_1 \in K$. We have that $B_1(x_1) \subseteq B_{d(x_0,x_1)+1}(x_0)$, which only contains b_n if $b < d(x_0, x_1) + 1$. Thus, $B_1(x_1)$ contains only finitely many points from $\mathscr{B}$, so x_1 cannot be a limit point of $\mathscr{B}$, a contradiction. This completes the proof. ■

3 Sequences

3.1 Introduction

Definition 3.1. Let S be a set. A sequence in S is a function $\mathbb{N} \to S$.

Notation. Let f be a sequence in S. We sometimes write $(f(n))_n$, $(f(n))_{n\in\mathbb{N}}$, or $(f(n))_{n=1}^{\infty}$.

Example 3.2. The function $f : \mathbb{N} \to \mathbb{N}$ defined by $f(k) = k + 3$ is a sequence.

Example 3.3. Let $\mathbb{N}^{\mathbb{N}}$ be the set of sequences of natural numbers, and let

$$g : \mathbb{N} \to \mathbb{N}^{\mathbb{N}}$$

be defined by

$$n \mapsto (k \mapsto k(n + 1)).$$

Then, g is the sequence $g_1, g_2, \ldots$, where $g_n = n + 1, 2n + 2, 3n + 3, \ldots$. We can write

$$g = ((k(n + 1))_k)_n\,.$$

3.2 Convergence

Definition 3.4. A sequence $(a_n)_n$ in a metric space (M, d) is said to converge to a point $L \in M$ if, for all $\varepsilon > 0$, there is some $N \in \mathbb{N}$ so that, when $n > N$, $d(n, L) < \varepsilon$. In this case, we call L the limit of $(a_n)_n$, and write

$$L = \lim_{n\to\infty} a_n.$$

If $(a_n)_n$ has no limit, it is said to diverge.

Example 3.5. The sequence $\left(\frac{1}{n}\right)_n$ has limit 0.

Example 3.6. In the discrete metric space, the sequence $(a_n)_n$ converges if and only if it is eventually constant.

Definition 3.7. Suppose that $(a_n)_n$ is a sequence in a metric space (M, d). Then, $(a_n)_n$ is said to be bounded if

$$\{a_n : n \in \mathbb{N}\}$$

is bounded.

Theorem 3.8. *Let (M, d) be a metric space and $(a_n)_n$ be a sequence in M. Then,*

(1) $(a_n)_n$ converges to L if and only if, for every open set $U \subseteq M$ with $L \in U$, all but finitely many terms of $(a_n)_n$ are in U.

(2) The limit of $(a_n)_n$ is unique if it exists.

(3) If $(a_n)_n$ converges, it is bounded.

(4) If $F \subseteq M$, and $x \in F'$, then there is a sequence in $F \setminus \{x\}$ converging to x.

Proof.

(1) Suppose that $\lim_{n\to\infty} a_n = L$, and let $U \subseteq M$ be open with $L \in U$. Since U is open, there is $\varepsilon > 0$ so that $B_\varepsilon(L) \subseteq U$. Since $(a_n)_n$ converges to L, there is some N so that, for every $n > N$, $d(n, L) < \varepsilon$. Then for such n, $a_n \in B_\varepsilon(L) \subseteq U$, so the only terms not in U are $a_1, a_2, \ldots, a_N$, of which there are only finitely many. Now suppose that, whenever $U \subseteq M$ is an open set and $L \in U$, U contains all but finitely many terms of $(a_n)_n$. Let $\varepsilon > 0$. Now, $B_\varepsilon(L)$ is open and contains L, so it contains all but finitely many terms of the sequence. Let $N = \max(n \in \mathbb{N} : a_n \notin B_\varepsilon(L))$. Then, if $n > N$, then $d(a_n, L) < \varepsilon$, meaning that it converges to L.

(2) Suppose that $(a_n)_n$ converges to L and $x \in M$ with $x \neq L$. Let $\varepsilon = \frac{d(L,x)}{2}$. Take N large enough that if $n > N$, $d(a_n, L) < \varepsilon$. This means that, if $n > N$, $d(a_n, x) > \frac{d(L,x)}{2}$. Thus, $B_\varepsilon(x)$ contains only finitely many terms of a_n, meaning that $(a_n)_n$ does not converge to x.

(3) Suppose that $\lim_{n\to\infty} a_n = L$. Let N be large enough that $n > N$, we have that $d(a_n, L) < 8$. Now, $\{a_1, a_2, \ldots, a_N, L\}$ is a finite set and so it is bounded. Let R be so that no two points in S are separated by more than R. Suppose that $n, m \in \mathbb{N}$, and consider $d(a_n, a_m) = r$. If $n, m \leq N$, $r < R$. If $n, m > N$, $r \leq d(a_n, L) + d(a_m, L) = 16$, and if exactly one is $\leq N$ (suppose n is), $r \leq d(a_n, L) + d(L, a_m) < 8 + R$. In any case, $r < 16 + R$, so the sequence is bounded.

(4) For each $n \in \mathbb{N}$, let $a_n \in B_{1/n}(x) \cap F \setminus \{x\}$. Consider the sequence $(a_n)_n$. For $\varepsilon > 0$, choose $N \in \mathbb{N}$ such that $\frac{1}{\varepsilon} < N$. Then, for $n > N$, we have that $d(a_n, x) < \frac{1}{n} < \frac{1}{N} < \varepsilon$, meaning that $(a_n)_n$ converges to x.

■

Proposition 3.9. *Suppose that (M, d) is a metric space, $(a_n)_n$ and $(y_n)_n$ are sequences in M, and $L \in M$. Suppose further that, for some T, we have that $\forall n > T, a_n = y_n$. Then $(a_n)_n$ converges to L if and only if $(y_n)_n$ does.*

Proof. Suppose $(a_n)_n$ converges to L. Let $\varepsilon > 0$. Since $(a_n)_n$ converges to L, there is some A so that if $k > A$, then $d(a_k, L) < \varepsilon$. Now, if $Y = \max(A, T)$ and $k > Y$, then $d(y_k, L) = d(a_k, L) < \varepsilon$. Thus, $(y_k)_k$ also converges to L. Exchanging the labels of $(a_n)_n$ and $(y_n)_n$ gives the other direction. ■

Theorem 3.10. *Suppose $(a_n)_n$ and $(b_k)_k$ are convergent sequences in $\mathbb{R}$ with limits a and b respectively. Let $c \in \mathbb{R}$. Then, the following are true.*

(1) $\lim_{n\to\infty} (a_n + b_n) = a + b$

(2) $\lim_{n\to\infty}(ca_n) = ca$

(3) $\lim_{n\to\infty}(a_nb_n) = ab$

(4) If $a \neq 0$*, then* $a_n = 0$ *only finitely often and* $\lim_{n\to\infty}\left(\frac{1}{a_n}\right) = \frac{1}{a}$

Proof.

(1) Let $\varepsilon > 0$. Choose A and B so that if $n > A$, $|a_n - a| < \frac{\varepsilon}{2}$, and if $n > B$, $|b_n - b| < \frac{\varepsilon}{2}$. Let $N = \max(A, B)$. Then, if $n > N$,

$$|a_n + b_n - (a+b)| = |(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence, $\lim_{n\to\infty}(a_n + b_n) = a + b$.

(2) Letting $c_n = c$, this result follows from (3).

(3) We want $|a_nb_n - ab| < \varepsilon$ for some $\varepsilon > 0$. We have that

$$|a_nb_n - ab| = |a_nb_n - a_nb + a_nb - ab| \leq |a_n||b_n - b| + |b||a_n - a|.$$

Because a_n converges, it must be bounded, so let $L = \max|a_n|$. If we let n_1 be so that for $n > n_1$, $|b_n - b| < \dfrac{\varepsilon}{2L}$ and n_2 be such that for $n > n_2$, $|a_n - a| < \dfrac{\varepsilon}{2|b|}$, then, for $n > \max(n_1, n_2)$,

$$|a_nb_n - ab| \leq |a_n||b_n - b| + |b||a_n - a| < L\frac{\varepsilon}{2L} + |b|\frac{\varepsilon}{2|b|} < \varepsilon.$$

However, this assumes that $L, |b| \neq 0$. If $L = 0$, then the theorem states that $\lim_{n\to\infty} 0 = 0$, which is trivial, and if $|b| = 0$, we may swap the roles of $(a_n)_n$ and $(b_n)_n$. Thus, the only problem arises when $|a| = |b| = 0$. In this case, we can assume that $|a_n|, |b_n| < 1$ for all n (as adding finitely many terms before it does not change the limit). Then, $|a_nb_n - ab| = |a_nb_n|$, which is smaller than $\varepsilon > 0$ for n such that $a_n < \varepsilon$. This is true of $n > N$ for some $N \in \mathbb{N}$.

(4) First, notice that if $a \neq 0$, then there is $N_0 \in \mathbb{N}$ so that if $n > N_0$ $|a - a_n| < \frac{|a|}{2}$. Then, $|a| \leq |a - a_n| + |a_n| < \frac{|a|}{2} + |a_n|$. Therefore, $|a_n| > \frac{|a|}{2} > 0$. In particular, $a_n \neq 0$ once $n > N_0$. Now, we want to control

$$\left|\frac{1}{a_n} - \frac{1}{a}\right| = \left|\frac{a - a_n}{aa_n}\right| \leq \frac{|a - a_n|}{|a||a|/2},$$

which we want to be less than $\varepsilon > 0$. Thus, we need $|a - a_n| < \dfrac{\varepsilon|a|^2}{2}$. Now, let N_1 be large enough that if $n > N_1$, we have $|a_n - a| < \dfrac{\varepsilon|a|^2}{2}$. Then, if $n > \max(N_1, N_0)$ (so that $a_n \neq 0$), we have that

$$\left|\frac{1}{a_n} - \frac{1}{a}\right| < \frac{2|a_n - a|}{|a|^2} < \varepsilon.$$

$\blacksquare$

Remark 3.11. If $(a_n + b_n)_n$ converges, it is not true that $(a_n)_n$ and $(b_n)_n$ converge.

Proposition 3.12. *Suppose that $(x_n)_n$ is a sequence in $\mathbb{R}^d$, Then $(x_n)_n$ converges to a limit $L \in \mathbb{R}^d$ if and only if the coordinate sequences $(x_n^{(j)})_n$ converge to the coordinates of L.*

Proof. Suppose that $(x_n)_n$ converges to L, and let $1 \leq j \leq d$. Let $\varepsilon > 0$. Take N large enough that if $n > N$, $|x_n - L| < \varepsilon$. Then, $\varepsilon > \sqrt{(x_n^{(1)} - L^{(1)})^2 + \cdots + (x_n^{(d)} - L^{(d)})^2} \geq \sqrt{(x_n^{(j)} - L^{(j)})^2} = |x_n^{(j)} - L^{(j)}|$. Hence, $\lim\limits_{n\to\infty} x_n^{(j)} = L^{(j)}$. Now, suppose that $\lim\limits_{n\to\infty} x_n^{(j)} = L^{(j)}$. Take $N_1, \ldots, N_d$ so that if $n > N_j$, then $|x_n^{(j)} - L^{(j)}| < \frac{\varepsilon}{\sqrt{d}}$. Then, if $n > \max(N_1, \ldots, N_d)$, we have that $|x_n - L| = \sqrt{(x_n^{(1)} - L^{(1)})^2 + \cdots + (x_n^{(d)} - L^{(d)})^2} < \sqrt{\frac{\varepsilon^2}{d} + \cdots + \frac{\varepsilon^2}{d}} = \varepsilon$. Hence, $\lim\limits_{n\to\infty} x_n = L$. $\blacksquare$

3.3 Subsequences

Definition 3.13. Suppose that $(a_n)_n$ is a sequence, and $n_1 < n_2 < \cdots$ is an increasing sequence of natural numbers. Then, $(a_{n_k})_k$ is said to be a subsequence of $(a_n)_n$.

Remark 3.14. Another point of view is if we let $a : \mathbb{N} \to X$ and $b : \mathbb{N} \to X$, then b is a subsequence of a if and only if there is some $n : \mathbb{N} \to \mathbb{N}$ so that $b = a \circ n$,

Proposition 3.15. *Every subsequence of a convergent sequence converges to the same limit as the original sequence.*

Proof. Let $(a_n)_n$ be a sequence converging to L, and let $\varepsilon > 0$. Then, there is some N_0 so that for $n > N_0$, $d(a_n, L) < \varepsilon$. Because the k_i's grow without bound, there must be some N_1 so that for $n > N_1$, $k_n > N_0$. Thus, for $n > N_1$, $d(a_{k_n}, L) < \varepsilon$, proving that $(a_{k_n})_n$ converges to L. $\blacksquare$

Theorem 3.16. *Suppose (K, d) is a compact metric space and $(x_n)_n$ is a sequence in K. Then, $(x_n)_n$ has a convergent subsequence.*

Proof. If $(x_n)_n$ has infinitely many terms that are equal, then it has a constant subsequence. Therefore, suppose that no term occurs more than finitely often. Then, $T = \{x_n : x \in \mathbb{N}\}$ is infinite, so T has a limit point $L \in K$. Let $k_0 = 0$. For each $n \in \mathbb{N}$, $B_{1/n}(L)$ contains infinitely many points in T. Then, we can find k_n so that $k_n > k_{n-1}$ and $x_{k_n} \in B_{1/n}(L)$. We claim $(x_{k_n})_n$ converges to L. Let $\varepsilon > 0$. Then, if $n > \frac{1}{\varepsilon}$, we have $x_{k_n} \in B_{1/n}(L)$ so $|x_{k_n} - L| < \frac{1}{n} < \varepsilon$. Hence, $\lim\limits_{n\to\infty} x_{k_n} = L$. $\blacksquare$

Corollary 3.17. *If (M, d) is a metric space and K is a compact subset, and $(x_n)_n$ is a sequence in K, then $(x_n)_n$ has a subsequence which converges to a limit in K.*

Corollary 3.18 (Bolzano-Weierstrass Theorem)**.** *Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.*

3.4 Cauchy Sequences

Definition 3.19. A sequence $(x_n)_n$ in a metric space (M,d) is said to be Cauchy if, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ so that whenever $n,k > N$, $d(x_n, x_k) < \varepsilon$.

Example 3.20. The sequence $(2^{-n})_n$ is Cauchy. To see this, note that if $\varepsilon > 0$ and N is large enough that $2^N > \varepsilon$, then for $n,k > N$,

$$\left|2^{-n} - 2^{-k}\right| = \left|\frac{2^k - 2^n}{2^{n+k}}\right| < \max\left(\frac{2^k}{2^{n+k}}, \frac{2^n}{2^{n+k}}\right) < 2^{-N} < \varepsilon.$$

Theorem 3.21. *Convergent sequences are Cauchy.*

Proof. Suppose that $(x_n)_n$ converges in a metric space (M,d). Now, let $\varepsilon > 0$. Because $(x_n)_n$ converges, there is some $N \in \mathbb{N}$ so that, for $n > N$, $d(x_n, \lim\limits_{n\to\infty} x_n) < \varepsilon/2$. Thus, for $n,k > N$, we have that $d(x_n, x_k) \le d(x_n, \lim\limits_{n\to\infty} x_n) + d(x_k, \lim\limits_{n\to\infty} x_n) < \varepsilon$. ■

Example 3.22. Suppose that $(q_n)_n$ is a sequence in $\mathbb{Q} \subset \mathbb{R}$ that converges to some $a \in \mathbb{R} \setminus \mathbb{Q}$. Then, $(q_n)_n$ is Cauchy, but not convergent in $\mathbb{Q}$.

Proposition 3.23. *Suppose that $(x_n)_n$ is a Cauchy sequence in the metric space (M,d) with a convergent subsequence $(x_{n_k})_k$. Then, $(x_n)_n$ converges (to the same limit).*

Proof. Let $L = \lim\limits_{k\to\infty} x_{n_k}$. Let $\varepsilon > 0$. Take K so that if $k > K$, $d(x_{n_k}, L) < \varepsilon/2$. Take N so that if $n,m > N$, $d(x_n, x_m) < \varepsilon/2$. Now let k be so large that $k > K$ and $n_k > N$. Now, for any $n > N$, $d(x_n, L) \le d(x_n, x_{n_k}) + d(x_{n_k}, L) < \varepsilon$. ■

Definition 3.24. A metric space is said to be complete if every Cauchy sequence converges.

Proposition 3.25. *If (K,d) is a compact metric space, it is complete.*

Proof. Since K is compact, any Cauchy sequence has a convergent subsequence, and so converges. ■

Lemma 3.26. *Cauchy sequences are bounded.*

Proof. Suppose $(x_n)_n$ is a Cauchy sequence in (M,d). Take N large enough that if $n,k > N$, $d(x_n, x_k) < 5$. Now, $\{x_1, \ldots, x_N, x_{n+1}\}$ is finite, and hence bounded. Thus, there is some $R > 0$ such that $\{x_1, \ldots, x_{N+1}\} \subseteq B_R(x_1)$. Now, if $n > N$, $d(x_n, x_1) \le d(x_n, x_{N+1}) + d(x_{N+1}, x_1) < 5 + R$, and if $n \le N$, $d(x_1, x_n) < R < 5 + R$. Hence, the $\{x_n : x \in \mathbb{N}\} \subseteq B_{5+r}(x_1)$, meaning it is bounded. ■

Corollary 3.27. *For any d, $\mathbb{R}^d$ is complete.*

Proof. Suppose that $(x_n)_n$ is Cauchy in $\mathbb{R}^d$. Then it is bounded. Take R large enough that the sequence is contained in $K = \underbrace{[-R,R] \times [-R,R] \times \cdots \times [-R,R]}_{d \text{ times}}$. This is compact, so $(x_n)_n$ must be convergent to a limit in $K \subseteq \mathbb{R}^d$. ■

Theorem 3.28. *Given a metric space (M,d), there is a metric space $(\overline{M}, \overline{d})$ so that*

- *$M \subseteq \overline{M}$ and $d|_{M\times M} = d$*
- *M is dense in $\overline{M}$*
- *$\overline{M}$ is complete.*

Furthermore, $\overline{M}$ is unique. $\overline{M}$ is called the completion of M.

Definition 3.29. Suppose that $(x_n)_n$ is a sequence in $\mathbb{R}$ (or any ordered set). Then $(x_n)_n$ is said to be (monotonically) increasing if $x_n \leq x_{n+1}$ for all n, and strictly (monotonically) increasing if $x_n < x_{n+1}$ for all n. Similarly for (monotonically) decreasing. It is said to be monotonic if it is either increasing or decreasing, and strictly monotonic if it is strictly increasing or strictly decreasing.

Theorem 3.30 (Monotone Convergence Theorem)**.** *A monotonic sequence in $\mathbb{R}$ converges if and only if it is bounded.*

Proof. We have already seen that convergent sequences are bounded. Now, suppose that $(x_n)_n$ is monotonic and bounded. If $(x_n)_n$ is decreasing and bounded, then $(-x_n)_n$ is increasing and bounded, and each converges if and only if the other does. Hence, we may assume $(x_n)_n$ is increasing. Let $L = \sup\{x_n : n \in \mathbb{N}\}$, and let $\varepsilon > 0$. Suppose that there is no $x_N \in B_\varepsilon(L)$. Then, $L - \varepsilon$ is an upper bound of $\{x_n : n \in \mathbb{N}\}$, which is a contradiction, so there must be some $N \in \mathbb{N}$ so that $x_N \in B_\varepsilon(L)$. We also have that $(x_n)_n$ is increasing, meaning that $L-\varepsilon < x_N \leq x_n \leq L$ for $n > N$. Thus, $|x_n - L| < \varepsilon$, completing the proof. ∎

Corollary 3.31. *If $(x_n)_n$ is a bounded monotonic sequence in $\mathbb{R}$, its limit is $\sup\{x_n : n \in \mathbb{N}\}$ if the sequence is increasing, and $-\sup\{-x_n : n \in \mathbb{N}\} = \inf\{x_n : n \in \mathbb{N}\}$ if the sequence is decreasing.*

Definition 3.32. Suppose $(x_n)_n$ is a sequence in the metric space (M, d). Then, $y \in M$ is a subsequential limit of $(x_n)_n$ if $(x_n)_n$ has a subsequence which converges to y.

Definition 3.33. Suppose $(x_n)_n$ is a sequence in $\mathbb{R}$. Its limit superior and limit inferior are defined as

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \sup\{x_k : k > n\}$$
$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} \inf\{x_k : k > n\},$$

provided these exist. If $(x_n)_n$ is not bounded above, we write $\limsup\limits_{n\to\infty} x_n = \infty$, and if it is not bounded below we write $\liminf\limits_{n\to\infty} x_n = -\infty$. If $(x_n)_n$ is bounded above but $\limsup\limits_{n\to\infty} x_n$ does not exist, we write $\limsup\limits_{n\to\infty} x_n = -\infty$, and if $(x_n)_n$ is bounded below, but $\liminf\limits_{n\to\infty} x_n$ does not exist, we write $\liminf\limits_{n\to\infty} x_n = \infty$.

Proposition 3.34. *Suppose that $(x_n)_n$ is a sequence in $\mathbb{R}$. If $(x_{n_k})_k$ is a convergent subsequence, then,*

$$\lim_{k\to\infty} x_{n_k} \leq \limsup_{n\to\infty} x_n.$$

Moreover, if $\limsup\limits_{n\to\infty} x_n$ is finite, there is a subsequence that converges to it.

Proof. Suppose that $(x_{n_k})_k$ is convergent. If $\limsup_{n\to\infty} = \infty$, we are done. Now, let $\varepsilon > 0$. Let K be large enough that if $k > K$,

$$|x_{n_k} - \lim_{q\to\infty} x_{n_q}| < \varepsilon.$$

In particular,

$$x_{n_k} > \lim_{q\to\infty} x_{n_q} - \varepsilon.$$

Now, for any n, there is some k with $n_k > n$, so $x_{n_k} \in \{x_m : m > n\}$. Thus, $\sup\{x_m : m > n\} \geq x_{n_k} > \lim_{q\to\infty} x_{n_q} - \varepsilon$. Then,

$$\lim_{n\to\infty} \sup\{x_m : m > n\} = \inf\{\sup\{x_m : m > n\}\} \geq \lim_{q\to\infty} x_{n_q} - \varepsilon.$$

However, ε was arbitrary, so

$$\lim_{n\to\infty} \sup\{x_m : m > n\} \geq \lim_{q\to\infty} x_{n_q}.$$

Now, let $L = \limsup_{n\to\infty} x_n$. Let $n_0 = 0$, and for $k \in \mathbb{N}$, there are infinitely many indices n for which $x_n > L - \frac{1}{k}$. Also, there are only finitely many indices where $x_n > L + \frac{1}{k}$. Then there are infinitely many indices where $L - \frac{1}{k} < x_n \leq L + \frac{1}{k}$. Let n_k be such that this is true, $n_k > n_{k-1}$. Now,

$$\lim_{k\to\infty} x_{n_k} = L,$$

as if $\varepsilon > 0$, if $k > 2\varepsilon^{-1}$, we have

$$|x_{n_q} - L| \leq \frac{1}{k} < \varepsilon,$$

for all $q > k$, so we are done. ■

4 Continuity

4.1 Limits

Definition 4.1. Suppose that (X, d_X) and (Y, d_Y) are metric spaces, $E \subseteq X$, $f : E \to Y$, and $p \in E'$ is a limit point of E. Then we say that a point $y \in Y$ is the limit of f at p provided

$$\forall\varepsilon > 0, \exists\delta > 0, \forall x \in B_\delta(p) \setminus \{p\}, d_Y(f(x), y) < \varepsilon.$$

Example 4.2. Let $S : \mathbb{R} \to \mathbb{R}$ with $|S(t)| \leq 1$. Then,

$$\lim_{t\to 0} tS(t^{-1}) = 0.$$

To this, let $\varepsilon > 0$, and notice that for $t \neq 0$,

$$|tS(t^{-1})| = |t||S(t^{-1})| \leq |t|.$$

Then, if $0 < |t - 0| < \varepsilon$, then

$$|tS(t^{-1}) - 0| \leq |t| < \varepsilon.$$

Theorem 4.3 (Sequential Characterization of Limits of Functions)**.** *Suppose that X and Y are metric spaces, $E \subseteq X$, $p \in E'$, and $f : E \to Y$. Then, $\lim_{x \to p} f(x) = y$ if and only if for every sequence $(x_n)_n$ in $E \setminus \{p\}$, with $\lim_{n\to\infty} x_n = p$, we have $\lim_{n\to\infty} f(x_n) = y$.*

Proof. Suppose that $\lim_{x\to p} f(x) = y$ and let $(x_n)_n$ be a sequence in $E \setminus \{p\}$ which converges to p. Let $\varepsilon > 0$. Choose $\delta > 0$ so that for $0 < d_X(x, p) < \delta$ we have that $d_Y(f(x), y) < \varepsilon$. Now, take N large enough so that if $n > N$, $0 < d_X(x_n, p) < \delta$. Then, for $n > N$, $d_Y(f(x_n), y) < \varepsilon$, meaning that $\lim_{n\to\infty} f(x_n) = y$. Now, suppose that f does not have limit y at p. Then there is some $\varepsilon > 0$ such that for any $\delta > 0$ there is $x \in E$ with $0 < d(x, p) < \delta$ and $d(f(x), y) \geq \varepsilon$. For each $n \in \mathbb{N}$, let $x_n \in E$ with $0 < d(x_n, p) < n^{-1}$ and $d(f(x_n), y) > \varepsilon$. Now, $\lim_{n\to\infty} x_n = p$. However, $(f(x_n))_n$ is never in the open set $B_\varepsilon(y)$, which contains y. Hence, $(f(x_n))_n$ does not converge to y. ∎

Corollary 4.4. *Suppose that X is a metric space, $E \subseteq X$, $p \in E'$, and $f, g : E \to \mathbb{R}$ with limits a and b respectively at p. Suppose also that $t \in \mathbb{R}$. Then,*

(1) $\lim_{x\to p}(f(x) + g(x)) = a + b$

(2) $\lim_{x\to p}(f(x)g(x)) = ab$

(3) If $a \neq 0$, then there is some $r > 0$ such that f is nonzero on $B_r(p) \setminus \{p\} \cap E$, and $\lim_{x\to p} \frac{1}{f(x)} = a^{-1}$.

Proof. These all follow from the Sequential Characterization of Limits of Functions, and previously proven results, once we show the first part of (3). To show this, note that if $a \neq 0$, then there is $\delta > 0$ so that if $x \in B_r(p) \setminus \{p\} \cap E$, we have $|f(x) - a| < |a|$, so $f(x) \neq 0$. ∎

4.2 Continuity

Definition 4.5. Suppose that X and Y are metric spaces, $f : X \to Y$, and $x_0 \in X$. Then f is said to be continuous at x_0 if, for every $\varepsilon > 0$, there is $\delta > 0$ so that if $x \in X$ with $d(x, x_0) < \delta$, we have $d(f(x), f(x_0)) < \varepsilon$. If f is continuous at x_0 for all $x_0 \in X$, then f is said to be continuous.

Example 4.6. Let $\pi_1 : \mathbb{R}^n \to \mathbb{R}$ defined by $\vec{x} \mapsto x_1$. Then, π_1 is continuous. To see this, let $\varepsilon > 0$, and use $\delta = \varepsilon$.

Example 4.7. The function $f : X \to X$ with $f(x) = x$ is continuous.

Proposition 4.8. *Suppose that X and Y are metric spaces, $f : X \to Y$, and $p \in X$ is not a limit point of X. Then f is continuous at p.*

Proof. Let $\varepsilon > 0$. Because p is not a limit point, there is some $\delta > 0$ so that $X \setminus \{p\} \cap B_\delta(p) = \varnothing$. Then, if $x \in X$ with $d_X(x, p) < \delta$, then $x = p$, so $d_Y(f(x), f(p)) = d(f(p), f(p)) = 0 < \varepsilon$. ∎

Theorem 4.9. *Suppose that X and Y are metric spaces with $f : X \to Y$ and $p \in X'$. Then, f is continuous at p is and only if $\lim_{x \to p} f(x) = f(p)$.*

Proof. Suppose that f is continuous at p. Let $\varepsilon > 0$. Choose $\delta > 0$ so that if $y \in X$ with $d_X(y, p) < \delta$ we have that $d_Y(f(y), f(p)) < \varepsilon$. Then, in particular, if $y \in X$ with $0 < d_X(y, p) < \delta$, we still have $d_Y(f(y), f(p)) < \varepsilon$ so $\lim_{y \to p} f(y) = f(p)$. Now suppose that $\lim_{x \to p} f(x) = f(p)$. Let $\varepsilon > 0$. Choose $\delta > 0$ so that if $0 < d_X(x, p) < \delta$ we have that $d_Y(f(x), f(p)) < \varepsilon$. But if $d(x, p) = 0$ then $x = p$, so $d_Y(f(x), f(p)) = 0 < \varepsilon$. Hence, we only need that $d_X(x, p) < \delta$. ■

Corollary 4.10. *Suppose that X is a metric space and $f, g : X \to \mathbb{R}$ are continuous at p. Then so are $f + g$, fg, and $\frac{f}{g}$ provided that $g(p) \neq 0$.*

Example 4.11. If $p \notin X'$, then continuity is automatic. Otherwise, it follows from the same properties of limits.

Corollary 4.12. *Polynomials are continuous.*

Proposition 4.13 (Sequential Characterisation of Continuity)**.** *Suppose that X and Y are metric spaces and $g : Y \to X$. Then g is continuous at $q \in Y$ if and only if for every sequence $(a_n)_n$ in Y which converges to q, $(g(a_n))_n$ converges to $g(q)$.*

Theorem 4.14. *Suppose that X and Y are metric spaces, and $f : X \to Y$. Then f is continuous if and only if whenever $y \subseteq Y$ is open, $f^{-1}(U)$ is also open.*

Proof. Suppose that f is continuous and $U \subseteq Y$ is open. We wish to show that $f^{-1}(U)$ is open. Let $x \in f^{-1}(U)$, so that $f(x) \in U$. Since U is open, there is some $\varepsilon > 0$ so that $B_\varepsilon(f(x)) \subseteq U$. Because f is continuous, there is some $\delta > 0$ so that if $t \in B_\delta(x)$, then $f(t) \in B_\varepsilon(f(x)) \subseteq U$, so $t \in f^{-1}(U)$. This is true for all $t \in B_\delta(x)$, so $B_\delta(x) \subseteq f^{-1}(U)$, so $f^{-1}(U)$ is open. Now, suppose that $f^{-1}(U)$ is open for all open sets $U \subseteq Y$. Let $x \in X$, and $\varepsilon > 0$. Now, $B_\varepsilon(f(x))$ is open, so $f^{-1}(B_\varepsilon(f(x)))$. Note that $x \in f^{-1}(B_\varepsilon(f(x)))$, since $f(x) \in B_\varepsilon(f(x))$, and so there is some $r > 0$ so that $B_r(x) \subseteq f^{-1}(B_\varepsilon(f(x)))$. Then, if $t \in X$ so that $t \in B_r(x)$, then $t \in f^{-1}(B_\varepsilon(f(x)))$, so that $f(t) \in B_\varepsilon(f(x))$. Thus, we may choose $\delta = r$, and we are done. ■

Corollary 4.15. *The composition of continuous functions is continuous.*

Proof. Suppose that X, Y, Z are metric spaces with $f : X \to Y$, and $g : Y \to Z$. Let $U \subseteq Z$ be open. Then, $g^{-1}(U) \subseteq Y$ is open. Thus, $f^{-1}(g^{-1}(U)) = (g \circ f)^{-1}(U) \subseteq X$ is open. Therefore, $g \circ f$ is continuous. ■

Theorem 4.16. *Suppose that X and Y are metric spaces, X is compact, and that $f : X \to Y$ is continuous. Then $f(X)$ is compact.*

Proof. Let $\mathcal{U}$ be an open cover of $f(X)$, and let $\mathcal{V} = \{f^{-1}(G) : g \in \mathcal{U}\}$. Because f is continuous, the elements of $\mathcal{V}$ are open sets. Suppose that $x \in X$. Then $f(x) \in f(X)$, so there is some $G \in \mathcal{U}$ with $f(x) \in G$, so $x \in f^{-1}(G)$. Thus, $\mathcal{V}$ forms an open cover of X. Then, it has a finite subcover $\{E_1, E_2, \ldots, E_n\}$. By construction, there are $G_1, G_2, \ldots, G_n \in U$ so that $E_i = f^{-1}(G_i)$. We claim that $\{G_1, G_2, \ldots, G_n\}$ covers $f(X)$, and so it is a finite subcover of $\mathcal{U}$. Let $y \in f(X)$. Then, $y = f(t)$ for some $t \in X$. Hence, $t \in E_i$ for some i. Thus, $t \in f^{-1}(G_i)$, so $f(t) = y \in G_i$. ■

Corollary 4.17 (Extreme Value Theorem)**.** *Suppose that $X \neq \varnothing$ is a compact metric space and $f : X \to \mathbb{R}$. Then there are $s, t \in X$ so that for any $x \in X$, $f(s) \leq f(x) \leq f(t)$.*

Proof. Because f is continuous, $f(X)$ is compact. In particular, it is closed and bounded. Then $\sup f(X)$ and $\inf f(X)$ exist in $\mathbb{R}$. Since $f(X)$ is closed, $\sup f(X), \inf f(X) \in f(X)$, and there are s, t as desired. ∎

Definition 4.18. Suppose that X and Y are metric spaces, and $f : X \to Y$. Then, f is uniformly continuous if, for every $\varepsilon > 0$, there is $\delta > 0$ so that for every $x, w \in X$ with $d_X(x, w) < \delta$, then $d_Y(f(x), f(w)) < \varepsilon$.

Theorem 4.19. *Suppose that K and X are metric spaces, K is compact, and $f : K \to X$ is continuous. Then f is uniformly continuous.*

Proof. Suppose that $f : K \to X$ is not uniformly continuous. Then there is some $\varepsilon > 0$ so that for every $n \in \mathbb{N}$ there are $x_n, y_n \in K$ so that $d(x_n, y_n) < n^{-1}$, but $d(f(x_n), f(y_n)) > \varepsilon$. Since K is compact, $(x_n)_n$ has a convergent subsequence $(x_{n_k})_k$ with limit $x \in K$. Now, $(y_{n_k})_k$ also converges to x. We claim that f is not continuous at x. For any $\delta > 0$, we may find some k so that $x_{n_k}, y_{n_k} \in B_\delta(x)$. However, $\varepsilon < d(f(x_{n_k}), f(y_{n_k})) \leq d(f(x_{n_k}), f(x)) + d(f(y_{n_k}), f(x))$, so at least one of these is greater than $\varepsilon/2$. Hence, f is not continuous at x. ∎

Proposition 4.20. *Suppose that $E \subseteq \mathbb{R}$ is not compact. Then there are continuous functions $f, g : E \to \mathbb{R}$ so that $f(E)$ is not bounded, and $g(E)$ is bounded, but has no maximum. If E is not closed, there is $h : E \to \mathbb{R}$ that is continuous but not uniformly continuous.*

Proof. Suppose that E is not bounded. Then, we may choose $f(x) = x$ so that $f(E)$ is not bounded. If E is not closed, let $x_0 \in E' \setminus E$. Then consider $f(x) = \frac{1}{x - x_0}$. If we want $f(x) > \varepsilon$ or $f(x) < -\varepsilon$, then we must choose x so that $|x - x_0| < \varepsilon^{-1}$. Such an x must exist, so $f(E)$ is not bounded. Now, if E is not bounded, then we may use $g(x) = \frac{x^2}{1+x^2}$. This is bounded above by 1, and below by 0. However, this has no maximum. If E is not closed, then consider $g(x) = \frac{1}{1+(x-x_0)^2}$. This is bounded below by 0, and it must be bounded above as it is impossible to make the denominator arbitrarily small. This is maximized when $x = x_0$, which is not possible. Now let $h(x) = \frac{1}{x - x_0}$. For $\varepsilon > 0$, we may choose $\delta = |\varepsilon(x_1 - x_0)(x - x_0)|$ to prove that $h(x)$ is continuous. However, we may make this as small as we like, so h is not uniformly continuous. ∎

Definition 4.21. Suppose that $f : X \to Y$ and Y is a metric space. Then f is bounded if $f(X)$ is bounded.

Definition 4.22. Suppose that X is a metric space, and $\varnothing \neq E \subseteq X$. Then E is disconnected if there are disjoint open sets $G_1, G_2 \subseteq X$ so that $E \subseteq G_1 \cup G_2$, and $E \cap G_i \neq \varnothing$ for $i = 1, 2$. If E is not disconnected, it is said to be connected. By convention, $\varnothing$ is disconnected.

Example 4.23. If $-\infty \leq a < b \leq \infty$, then (a, b) is connected. Suppose that (a, b) is disconnected. Then there are open sets $G_1, G_2 \subseteq \mathbb{R}$ with $G_1 \cap G_2 = \varnothing$, $G_1 \cap (a, b) \neq \varnothing$, $G_2 \cap (a, b) \neq \varnothing$, $(a, b) \subseteq G_1 \cup G_2$. Let $x \in G_1 \cap (a, b)$, and $y \in G_2 \cap (a, b)$. Suppose without loss of generality that $x < y$. Note that $G_1 \cap (-\infty, y)$ is non-empty and bounded above, so $T = \sup(G_1 \cap (-\infty, y))$ exists. Note also that $a < x \leq T \leq y < b$, so $T \in (a, b)$. Now, $T \in \overline{(G_1 \cap (-\infty, y))} \subseteq \overline{G_1} \subseteq \overline{G_2^c} = G_2^c$. Note that, if $T_\in G_1$, then since G_1 is open, for some $r > 0$, we have $(T - r, T + r) \subseteq G_1$. However, this can only happen if $T = y$, so $T \in G_2$. This means that $T = y \in G_1 \cap G_2$, which is a contradiction.

Theorem 4.24. *Suppose that X and Y are metric spaces, $f : X \to Y$ is continuous, and $E \subseteq X$ is connected. Then $f(E)$ is also connected.*

Proof. Suppose that $f(E)$ is disconnected. Take $G_1, G_2 \subseteq Y$ satisfying the conditions of disconnectedness. Then, $f^{-1}(G_1)$ and $f^{-1}(G_2)$ are open. Note also that if $x \in X$, and $f(x) \in G_i$, then $x \in f^{-1}(G_i)$. Finally, if $x \in f^{-1}(G_1) \cap f^{-1}(G_2)$, then $f(x) \in G_1 \cap G_2$, so $f^{-1}(G_1) \cap f^{-1}(G_2) = \varnothing$. Thus, E is disconnected, a contradiction. ∎

Corollary 4.25 (Intermediate Value Theorem)**.** *Suppose that $f : [a, b] \to \mathbb{R}$ is continuous and y is between $f(a)$ and $f(b)$. Then there is $c \in [a, b]$ so that $f(c) = y$.*

Proof. Suppose that $y \notin f([a, b])$. Let $G_1 = (-\infty, y)$ and $G_2 = (y, \infty)$. Then, $G_1 \cap G_2 \neq \varnothing$, $f([a, b]) \subseteq \mathbb{R} \setminus \{y\} = G_1 \cup G_2$, and $f([a, b]) \cap G_i$ contains either $f(a)$ or $f(b)$. Hence, $[a, b]$ is disconnected. However, let G_1, G_2 be a disconnection of $[a, b]$. Since (a, b) is connected, either $(a, b) \subseteq G_1$ or $(a, b) \subseteq G_2$. Assume that this is true for G_1. Then either $a \in G_2$,or $b \in G_2$. If $a \in G_2$, since it is open, for $r > 0$ small enough $(a - r, a + r) \subseteq G_2$ so $(a, a + r) \subseteq G_1 \cap G_2$, a contradiction. Similarly for b. ∎

4.3 The Extended Real Numbers

Definition 4.26. The extended real numbers are the set $\mathbb{R}_\infty = \mathbb{R} \cup \{-\infty, \infty\}$. We define arithmetic in the expected way. We say also that $-\infty < x \in \mathbb{R} < \infty$.

Remark 4.27. If $S \subseteq \mathbb{R}$ not bounded above (below) in $\mathbb{R}$, then in $\mathbb{R}_\infty$, $\sup S$ ($\inf S$) is ∞ ($-\infty$). Also, in $\mathbb{R}_\infty$, $\sup \varnothing = -\infty$, and $\inf \varnothing = \infty$.

Definition 4.28. We say that a sequence $(a_n)_n$ in $\mathbb{R}$ (or $\mathbb{R}_\infty$) diverges to ∞ if for every $T < \infty$, there is $N \in \mathbb{N}$ so that for all $n > N$, $a_n > T$. In this case we write $\lim\limits_{n \to \infty} a_n = \infty$. We likewise say that $(a_n)_n$ diverges to $-\infty$ if for all $T > -\infty$, there is $N \in \mathbb{N}$ so that if $n > N$, $a_n < T$. We extend these ideas to functions using Sequential Characterization of Limits.

Definition 4.29. Suppose that X is a metric space, $E \subseteq \mathbb{R}$ is not bounded above, $L \in X$, and $f : E \to X$. Then, $\lim\limits_{t \to \infty} f(t) = L$ provided for every $\varepsilon > 0$, there is T so that for all $t \in E$ with $t > T$, $d_X(f(t), L) < \varepsilon$.

5 Calculus

5.1 Differential

Definition 5.1. Suppose $a, b \in \mathbb{R}$ with $a < b$, and $f : (a, b) \to \mathbb{R}$. Then f is said to be differentiable at a point $x \in (a, b)$ if the difference quotient

$$\frac{f(y) - f(x)}{y - x}$$

has a limit as y approaches x. In this case, we write

$$f'(x) = \lim_{y \to x} \frac{f(y) - f(x)}{y - x}.$$

Definition 5.2. Let $a, b \in \mathbb{R}$ with $a < b$. If $f : (a, b) \to \mathbb{R}$ is differentiable at every point in (a, b), it is said to be differentiable, and its derivative is the function

$$x \mapsto f'(x) = \lim_{y \to x} \frac{f(y) - f(x)}{y - x}.$$

Example 5.3. Constants are differentiable. Suppose that $f : \mathbb{R} \to \mathbb{R}$ with $f(t) = C$. Then,

$$\lim_{y \to x} \frac{f(y) - f(x)}{y - x} = \lim_{y \to x} \frac{C - C}{y - x} = 0.$$

Example 5.4. If $g : \mathbb{R} \to \mathbb{R}$ with $g(t) = t$, then g is differentiable. To see this, note that

$$\lim_{y \to x} \frac{g(y) - g(x)}{y - x} = \lim_{y \to x} \frac{y - x}{y - x} = 1.$$

Example 5.5. If $h : \mathbb{R} \to \mathbb{R}$ with $h(t) = |t|$, then it is not differentiable (at 0). We must show that $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ with $t \mapsto |t|/t$ has no limit at 0. Note that, for example,

$$\lim_{n \to \infty} \frac{|1/n|}{1/n} = 1 \neq -1 = \lim_{n \to \infty} \frac{|-1/n|}{-1/n}.$$

Theorem 5.6. *If $f : (a, b) \to \mathbb{R}$ is differentiable at $x \in (a, b)$, then f is continuous there.*

Proof. It is clear that $0 = (f'(x)) \lim_{t \to x}(t - x)$. This is equal to

$$\lim_{t \to x} \frac{f(t) - f(x)}{t - x} \lim_{t \to x} t - x$$

$$= \lim_{t \to x} f(t) - f(x) = \lim_{t \to x}(f(t)) - f(x).$$

Thus, $\lim_{t \to x} f(t)$ exists and is equal to $f(x)$, so f is continuous. ■

Theorem 5.7. *Suppose that $f, g : (a, b) \to \mathbb{R}$ are differentiable at $x \in (a, b)$. Then so are $f + g$, $f \cdot g$, and*

$$(f + g)'(x) = f'(x) + g'(x)$$

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Proof. We have that

$$\begin{aligned}
f'(x) + g'(x) &= \lim_{y \to x} \frac{f(y) - f(x)}{y - x} + \lim_{y \to x} \frac{g(y) - g(x)}{y - x} \\
&= \lim_{y \to x} \left(\frac{f(y) - f(x)}{y - x} + \frac{g(y) - g(x)}{y - x} \right) \\
&= \lim_{y \to x} \left(\frac{(f(y) + g(y)) - (f(x) + g(x))}{y - x} \right) \\
&= (f + g)'(x).
\end{aligned}$$

We also have that

$$\begin{aligned}
f'(x)g(x) + f(x)g'(x) &= \lim_{y \to x} \frac{f(y) - f(x)}{y - x} g(x) + \lim_{y \to x} f(y) \lim_{y \to x} \frac{g(y) - g(x)}{y - x} \\
&= \lim_{y \to x} \frac{f(y)g(x) - f(x)g(x) + f(y)g(y) - f(y)g(y)}{y - x} \\
&= \lim_{y \to x} \frac{f(y)g(y) - f(x)g(x)}{y - x} \\
&= (fg)'(x).
\end{aligned}$$

■

Corollary 5.8. *Polynomials are differentiable.*

Proposition 5.9. *Suppose that $f : (a, b) \to \mathbb{R}$, $x(a, b)$. Then f is differentiable at x if and only if there is a constant C and a function $E : (a, b) \to \mathbb{R}$ so that $\lim_{t \to x} E(t) = 0$, $E(x) = 0$, and $f(t) = f(x) + C(t - x) + E(t)(t - x)$. In this case, $C = f'(x)$.*

Proof. Suppose that such an E and C exist. Then,

$$\lim_{y \to x} \frac{f(y) - f(x)}{y - x} = \lim_{y \to x} \frac{(f(x) + C(y - x) + E(y)(y - x)) - f(x)}{y - x}$$

$$= \lim_{y \to x} \frac{C(y - x) + E(y)(y - x)}{y - x} = \lim_{y \to x} C + E(y) = C.$$

Thus, the limit exists and $f'(x) = C$. Now, let $C = f'(x)$ and define $E(t) = (t - x)^{-1}(f(t) - f(x) - C(t - x))$ for $t \neq x$ and $E(x) = 0$. Note that

$$\lim_{t \to x} E(t) = \lim_{t \to x} \frac{f(t) - f(x)}{t - x} - C = C - C = 0.$$

Note also that $f(t) = f(x) + C(t - x) + E(t)(t - x)$. ■

Theorem 5.10. *Suppose that $f : (a, b) \to (c, d) \subseteq \mathbb{R}$, $g : (c, d) \to \mathbb{R}$, f is differentiable at $x \in (c, d)$, and that g is differentiable at $f(x)$. Then $(g \circ f)$ is differentiable at x, and $(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$.*

Proof. We know that there are functions $E_1 : (a, b) \to \mathbb{R}$ and $E_2 : (c, d) \to \mathbb{R}$ with $\lim_{t \to x} E_1(t) = 0$, $\lim_{y \to f(x)} E_2(t) = 0$, and

$$f(t) = f(x) + f'(x)(t - x) + E_1(t)(t - x)$$
$$g(y) = g(f(x)) + g'(f(x))(y - f(x)) + E_2(y)(y - f(x)).$$

Thus,

$$g(f(t)) = g(f(x)) + g'(f(x))(f(t) - f(x)) + E_2(f(t))(f(t) - f(x))$$
$$= g(f(x)) + g'(f(x))(f'(x)(t - x) + E_1(t)(t - x)) + E_2(f(t))(f'(x)(t - x) + E_1(t)(t - x)).$$

It therefore suffices to show that

$$\lim_{t \to x} g'(f(x))E_1(t) + E_2(f(t))f'(x) + E_2(f(t))E_1(t) = 0.$$

To do so, it is enough to show

$$\lim_{t \to x} E_2(f(t)) = 0.$$

Let $\varepsilon > 0$. Since $\lim_{y \to f(x)} E_2(y) = 0 = E_2(f(x))$, there is $\delta > 0$ so that if $0 < |y - f(x)| < \delta$, $|E_2(y)| < \varepsilon$. Since f is differentiable (and so continuous) at x, there is $\rho > 0$ so that if $|t - x| < \rho$ we have $|f(t) - f(x)| < \delta$, and so $|E_2(f(t))| < \varepsilon$. Hence, $\lim_{t \to x} E_2(f(t)) = 0$. Thus, $g \circ f'(x) = g'(f(x))f'(x)$. ■

Example 5.11. Let us look to understand $(f^{-1})'$ given $f : (a, b) \to \mathbb{R}$. Suppose that $x \in (a, b)$, and there is some $\delta > 0$ so that f is injective on $(x - \delta, x + \delta)$ with $f((x - \delta, x + \delta)) = S$. Then, $f^{-1} : S \to (x - \delta, x + \delta)$. Now, $(f^{-1} \circ f)(y) = y$ for all $y \in (x - \delta, x + \delta)$. Thus, if f is differentiable at x and f^{-1} is going to be differentiable at $f(x)$, we have the following.

$$x = (f^{-1} \circ f)(x)$$
$$1 = (f^{-1} \circ f)'(x)$$
$$= (f^{-1})'(f(x)) \cdot f'(x).$$

Thus, if $f'(x) \neq 0$,

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)}.$$

In general,

$$(f^{-1})'(t) = \frac{1}{f'(f^{-1}(t))},$$

provided that $f^{-1}(t)$ satisfies the same assumptions as x.

Lemma 5.12. *Let $q : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be defined by $t \mapsto t^{-1}$. Then q is differentiable and $q'(t) = -t^{-2}$.*

Proof. Note that for $x, t \in \mathbb{R} \setminus \{0\}$,

$$\frac{\frac{1}{t} - \frac{1}{x}}{t - x} = \frac{-1}{tx}.$$

Since $t \mapsto (t^{-1})(-x^{-1})$ is continuous, we have $q'(x) = \lim_{t \to x}(t^{-1})(-x^{-1}) = -x^{-2}$. ■

Theorem 5.13. *Suppose $f, g : (a, b) \to \mathbb{R}$ are differentiable at $x \in (a, b)$, and $g(x) \neq 0$. Then $\frac{f}{g}$ is defined on a set containing an open interval containing x, is differentiable at x, and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.$$

Proof. First let's check that the quotient is defined on an open set containing x. Since g is differentiable at x, it is continuous there. Then there is some $\delta > 0$ so that if $y \in (a, b)$ with $|x - y| < \delta$ then $|g(x) - g(y)| < |g(x)|/2$. Then $|g(y)| > |g(x)|/2 > 0$. Hence, f/g is defined at least on $(a, b) \cap (x - \delta, x + \delta)$. Now, if $q(t) = t^{-1}$, then $(f/g)(t) = f(t)q(g(t))$. Hence $(f/g)'(t) = f'(x)q(g(x)) + f(x)q'(g(x))g'(x)$. This is equal to the statement of the theorem. ■

Definition 5.14. Suppose X is a metric space, S is an ordered set, and $f : X \to S$. A point $x \in X$ is called a local maximum (local minimum respectively) if there is an open set $U \subseteq X$ with $x \in U$ so that for all $y \in U$, we have $f(y) \leq f(x)$ ($f(y) \geq f(x)$ respectively).

Proposition 5.15. *Suppose that $f : (a, b) \to \mathbb{R}$ with a local extremum (local minimum or maximum) at $x \in (a, b)$. Then, if f is differentiable there, $f'(x) = 0$.*

Proof. Let us suppose that $f'(x)$ exists but is nonzero. Then, there is $\delta > 0$ so that if $y \in (a, b)$ with $|x - y| < \delta$,

$$\left|\frac{f(y) - f(x)}{y - x} - f'(x)\right| < \frac{|f'(x)|}{2}.$$

Now, we have the following

$$\begin{aligned}
|f'(x)| &\leq \left|f'(x) - \frac{f(y) - f(x)}{y - x}\right| + \left|\frac{f(y) - f(x)}{y - x}\right| \\
|f'(x)| - \left|\frac{f(y) - f(x)}{y - x} - f'(x)\right| &\leq \left|\frac{f(y) - f(x)}{y - x}\right| \\
|f'(x)| - \frac{|f'(x)|}{2} &\leq \left|\frac{f(y) - f(x)}{y - x}\right| \\
\frac{|f'(x)|}{2} &\leq \left|\frac{f(y) - f(x)}{y - x}\right|.
\end{aligned}$$

Now, let $0 < \varepsilon < \delta$. Then,

$$\left|\frac{f(x - \varepsilon) - f(x)}{-\varepsilon}\right| > 0,$$

and

$$\left|\frac{f(x+\varepsilon)-f(x)}{\varepsilon}\right| > 0.$$

Since both are between $|f'(x)|/2$ of $f'(x) \neq 0$, either all 3 are positive (in which case $f(x-\varepsilon) < f(x) < f(x+\varepsilon)$), or all 3 are negative (in which case $f(x+\varepsilon) < f(x) < f(x-\varepsilon)$). However, this means that there is no open interval where f attains an extreme value at x. ■

Corollary 5.16. *If $f : [a, b] \to \mathbb{R}$ is continuous and differentiable on (a, b), then the global extrema occur either at a, b, or at points where f' vanishes.*

Proposition 5.17. *Suppose $f, g : [a, b] \to \mathbb{R}$ are continuous and differentiable on (a, b). Then there is some $x \in (a, b)$ so that*

$$((f(b) - f(a))g'(x) = (g(b) - g(a))f'(x).$$

Proof. Let $h : [a, b] \to \mathbb{R}$ be defined by

$$t \mapsto (f(b) - f(a))g(t) - (g(b) - g(a))f(t).$$

Then, h is continuous on $[a, b]$ and differentiable on (a, b), since f and g are. Note that $h(a) = h(b) = 0$. If h is constant, then $h'(t) = 0$, so the equality holds for all $x \in (a, b)$. If h is nonzero, then it either has a nonzero global minimum, or a nonzero global maximum. Let the nonzero global extremum occur at $x \in (a, b)$. Then, x is a local extremum as well, so $h'(x) = 0$, so the desired equality holds at x. ■

Corollary 5.18 (Mean Value Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous and differentiable on (a, b). Then there is some $c \in (a, b)$ so that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Let $g(x) = x$ in the previous Proposition. Then there is some $c \in (a, b)$ with $f(b) - f(a) = (b - a)f'(c)$, so $f'(c) = \frac{f(b)-f(a)}{b-a}$. ■

Corollary 5.19. *Suppose $f : [a, b] \to \mathbb{R}$ is continuous and differentiable on (a, b). Then,*

- *If $f'(t) \geq 0$ for all $t \in (a, b)$, then f is monotonically increasing.*
- *If $f'(t) \leq 0$ for all $t \in (a, b)$, then f is monotonically decreasing.*
- *If $f'(t) = 0$ for all $t \in (a, b)$, then f is constant.*

Proof. Suppose $a \leq s < t \leq b$. Applying the Mean Value Theorem to f on $[s, t]$, we find $x \in (s, t)$ so that $f(t) - f(s) = f'(x)(t - s)$. Hence, if $f'(x) \geq 0$ for all $x \in (s, t) \subseteq (a, b)$, we have $f(t) - f(s) \geq 0$, if $f'(x) \leq 0$ for all $x \in (s, t) \subseteq (a, b)$, we have $f(t) - f(s) \leq 0$, and if $f'(x) = 0$ for all $x \in (s, t) \subseteq (a, b)$ then $f(t) - f(s) = 0$. ■

5.2 Integral

Definition 5.20. A partition of an interval $[a, b]$ is a finite set of points in $[a, b]$ which includes a and b. Given a partition $\mathcal{P}$, we write $\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_n = b\}$ to assign the labels $x_0, x_1, \ldots, x_n$ to the points in $\mathcal{P}$ in increasing order. In this case, we denote $\Delta x_i = x_i - x_{i-1}$.

Definition 5.21. Let $\mathcal{P}$ be a partition of $[a, b]$. We say a partition $\mathcal{Q}$ is a refinement of $\mathcal{P}$ if $\mathcal{P} \subseteq \mathcal{Q}$.

Lemma 5.22. *If $\mathcal{P}_1$ and $\mathcal{P}_2$ are partitions of $[a, b]$, they have a common refinement.*

Proof. Take $\mathcal{P}_1 \cup \mathcal{P}_2$. ■

Notation. Given a set S and a function $\phi : S \to \mathbb{R}$ (or any ordered set). We write $\sup\limits_{x \in S} \phi(x)$ for $\sup\{\phi(x) : x \in S\}$.

Definition 5.23. Suppose $f : [a, b] \to \mathbb{R}$ is bounded and $\mathcal{P}$ is a partition of $[a, b]$. Write $\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_n = b\}$ We define

$$U(f, \mathcal{P}) = \sum_{i=1}^{n} \sup_{x \in [x_{i-1}, x_i]} f(x) \Delta x_i,$$

and

$$L(f, \mathcal{P}) = \sum_{i=1}^{n} \inf_{x \in [x_{i-1}, x_i]} f(x) \Delta x_i.$$

We abbreviate these as

$$U(f, \mathcal{P}) = \sum_{i=1}^{n} M_i \Delta x_i \qquad \text{and} \qquad L(f, \mathcal{P}) = \sum_{i=1}^{n} m_i \Delta x_i,$$

but only when f and $\mathcal{P}$ are unambiguous.

Definition 5.24. We define the upper and lower Darboux integrals of f as

$$U(f) = \inf_{\mathcal{P}} U(f, \mathcal{P})$$

and

$$L(f) = \sup_{\mathcal{P}} L(f, \mathcal{P}),$$

where the inf and sup are taken over the set of partitions of $[a, b]$. If $U(f) = L(f)$ we say f is integrable and write

$$\int_a^b f(t)dt = U(f) = L(f).$$

Proposition 5.25. *Suppose $f : [a, b] \to \mathbb{R}$, and $\mathcal{P}$, $\mathcal{Q}$ are partitions of $[a, b]$ so that $\mathcal{P}$ refines $\mathcal{Q}$. Then, $L(f, \mathcal{Q}) \leq L(f, \mathcal{P})$ and $U(f, \mathcal{Q}) \geq U(f, \mathcal{P})$*

Proof. Suppose that

$$\mathcal{Q} = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

Then,

$$U(f, \mathcal{Q}) = \sum_{i=1}^{n} \sup_{x \in [x_{i-1}, x_i]} f(x)(x_i - x_{i-1}).$$

Set $\mathcal{P}_i = \mathcal{P} \cap [x_{i-1}, x_i]$, and let $f_i = f|_{[x_{i-1}, x_i]}$. Now, writing $\mathcal{P} = \{a = y_0 < y_1 < \cdots < y_m = b\}$,

$$\begin{aligned} U(f, \mathcal{P}) &= \sum_{i=1}^{m} \sup_{x \in [y_{i-1}, y_i]} f(x)(y_i - y_{i-1}) \\ &= \sum_{j=1}^{n} \left(\sum_{i : y_i \in \mathcal{P}_j} \sup_{x \in [y_{i-1}, y_i]} f(x)(y_i - y_{i-1}) \right). \end{aligned}$$

Let $\mathcal{P}_j = \{x_{j-1} = z_{j,0} < z_{j,1} < \cdots < z_{j,n_j} = x_j\}$. Then, this inner sum is

$$\sum_{i=1}^{n_j} \sup_{x \in [z_{j,i-1}, z_{j,i}]} f(x)(z_{j,i} - z_{j,i-1}) \leq \sum_{i=1}^{n_j} \sup_{x \in [x_{j-1}, x_j]} f(x)(z_{j,i} - z_{j,i-1}).$$

Thus, the entire sum is less than or equal to

$$\sum_{j=1}^{n} \sum_{i=1}^{n_j} \sup_{x \in [x_{j-1}, x_j]} f(x)(z_{j,i} - z_{j,i-1}) = \sum_{j=1}^{n} \sup_{x \in [x_{j-1}, x_j]} \sum_{i=1}^{n_j} f(x)(z_{j,i} - z_{j,i-1})$$

$$= \sum_{j=1}^{n} \sup_{x \in [x_{j-1}, x_j]} f(x)(z_{j,n_j} - z_{j,0}) = \sum_{j=1}^{n} \sup_{x \in [x_{j-1}, x_j]} f(x)(x_j - x_{j-1}) = U(f, \mathcal{Q}).$$

The proof for $L(f, \mathcal{P}) \geq L(f, \mathcal{Q})$ is the same but replacing sups with infs. Finally,

$$\begin{aligned} L(f, \mathcal{P}) &= \sum_{i=1}^{m} \inf_{x \in [y_{i-1}, y_i]} f(x)(y_i - y_{i-1}) \\ &\leq \sum_{i=1}^{m} \sup_{x \in [y_{i-1}, y_i]} f(x)(y_i - y_{i-1}) \\ &= U(f, \mathcal{P}). \end{aligned}$$

■

Corollary 5.26. *For any partitions $\mathcal{P}$, $\mathcal{Q}$, we have*

$$L(f, \mathcal{P}) \leq U(f, \mathcal{Q}).$$

Proof. We have that $L(f, \mathcal{P}) \leq L(f, \mathcal{P} \cup \mathcal{Q}) \leq U(f, \mathcal{P} \cup \mathcal{Q}) \leq U(f, \mathcal{Q})$. ∎

Example 5.27. Let $f : [0, 1] \to \mathbb{R}$ be defined by

$$t \mapsto \begin{cases} 1 & t \in \mathbb{Q} \\ 0 & t \notin \mathbb{Q} \end{cases}.$$

For any partition $\mathcal{P} = \{0 = x_0 < x_1 < \cdots < x_n = 1\}$, there are rationals $q_1, q_2, \ldots, q_m$ and irrationals $t_1, t_2, \ldots, t_n$ so that $x_{j-1} < q_j < x_j$ and $x_{j-1} < t_j < x_j$. Then, $1 = \sup_{x \in [0,1]} f(x) \geq \sup_{x \in [x_{j-1}, x_j]} f(x) \geq f(q_j) = 1$, and $0 = f(t_j) \geq \inf_{x \in [x_{j-1}, x_j]} f(x) \geq \inf_{x \in [0,1]} f(x) = 0$. So,

$$L(f, \mathcal{P}) = \sum_{j=1}^{n} \inf_{x \in [x_{j-1}, x_j]} f(x)(x_j - x_{j-1}) = \sum 0(x_j - x_{j-1}) = 0.$$

Similarly, $U(f, \mathcal{P}) = 1$. Thus, f is not integrable.

Proposition 5.28. *Suppose $f : [a, b] \to \mathbb{R}$. Then, $L(f) \leq U(f)$.*

Proof. For any partitions $\mathcal{P}$ and $\mathcal{Q}$,

$$L(f, \mathcal{P}) \leq U(f, \mathcal{Q}).$$

Thus, taking infimums over $\mathcal{Q}$, $L(f, \mathcal{P}) \leq U(f)$. Taking supremums over $\mathcal{P}$, $L(f) \leq U(f)$. ∎

Remark 5.29. Suppose $f : [a, b] \to [m, M]$. Then, for any partition $\mathcal{P}$, we have $\mathcal{P}_0 = \{a, b\} \subseteq \mathcal{P}$. Thus, $\inf_{x \in [a,b]} f(x)(b - a) = L(f, \mathcal{P}_0) \leq L(f, \mathcal{P}) \leq L(f)$, and $\sup_{x \in [a,b]} f(x)M(b - a) = U(f, \mathcal{P}_0) \geq U(f, \mathcal{P}) \geq U(f)$.

Proposition 5.30. *Suppose $f : [a, b] \to \mathbb{R}$ is bounded. Then f is integrable if and only if, for all $\varepsilon > 0$, there is a partition $\mathcal{P}$ of $[a, b]$ so that $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$.*

Proof. Let $\varepsilon > 0$ and suppose that f is integrable. Choose partitions $\mathcal{P}$, $\mathcal{Q}$ of $[a, b]$ so that $U(f, \mathcal{P}) < U(f) + \varepsilon/2$ and $L(f, \mathcal{Q}) > L(f) - \varepsilon/2$. Then,

$$L(f) - \varepsilon/2 < L(f, \mathcal{Q}) \leq L(f, \mathcal{P} \cup \mathcal{Q}) \leq U(f, \mathcal{P} \cup \mathcal{Q}) \leq U(f, \mathcal{P}) < U(f) + \varepsilon/2.$$

Thus,

$$U(f, \mathcal{P} \cup \mathcal{Q}) - L(f, \mathcal{P} \cup \mathcal{Q}) < U(f) + \varepsilon/2 - (L(f) - \varepsilon/2) = \varepsilon.$$

Now suppose that $L(f) \neq U(f)$. Then $L(f) < U(f)$. For any partition $\mathcal{P}$, $U(f, \mathcal{P}) - L(f, \mathcal{P}) \geq U(f) - L(f, \mathcal{P}) \geq U(f) - L(f) > 0$. If we choose $\varepsilon = U(f) - L(f)$, then there is no partition $\mathcal{P}$, so that $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$. ∎

Proposition 5.31. *Continuous functions are integrable.*

Proof. Suppose $f : [a,b] \to \mathbb{R}$ is continuous. Since $[a,b]$ is compact, f is uniformly continuous. Let $\varepsilon > 0$. Take $\delta > 0$ so that if $x, y \in [a,b]$ with $|x-y| < \delta$, we have $|f(x) - f(y)| < \varepsilon/(b-a)$. Take a partition $\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_n = b\}$ so that $x_i - x_{i-1} < \delta$ for each $i = 1, 2, \ldots, n$. Since f is continuous on $[x_{i-1}, x_i]$, there are $s_i, t_i \in [x_{i-1}, x_i]$ so that

$$f(s_i) = \inf_{x \in [x_{i-1}, x_i]} f(x) \qquad f(t_i) = \sup_{x \in [x_{i-1}, x_i} f(x).$$

But $|s_i - t_i| < \delta$, so $|f(s_i) - f(t_i)| < \varepsilon/(b-a)$. Now

$$\begin{aligned} U(f,\mathcal{P}) - L(f,\mathcal{P}) &= \sum_{i=1}^{n} \sup_{x \in [x_{i-1}, x_i]} f(x)(x_i - x_{i-1}) - \sum_{i=1}^{n} \inf_{x \in [x_{i-1}, x_i]} f(x)(x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} (f(t_i) - f(s_i))(x_i - x_{i-1}) \\ &< \sum_{i=1}^{n} \frac{\varepsilon}{b-a}(x_i - x_{i-1}) \\ &= \frac{\varepsilon}{b-a}(b-a) = \varepsilon. \end{aligned}$$

Thus, f is integrable. ∎

Proposition 5.32. *Suppose f, g are integrable on $[a,b]$, $a < c < b$, and $\lambda \in \mathbb{R}$. Then,*

(1) $f + \lambda g$ is integrable and $\int_a^b f(t) + \lambda g(t) dt = \int_a^b f(t)dt + \lambda \int_a^b g(t)dt$.

(2) If $f \geq 0$, then $\int_a^b f(t)dt \geq 0$.

(3) $\int_a^b \lambda dt = \lambda(b-a)$.

(4) f is integrable on $[a,c]$ and $[c,b]$, and $\int_a^b f(t)dt = \int_a^c f(t)dt + \int_c^b f(t)dt$.

(5) $\left|\int_a^b f(t)dt\right| \leq \int_a^b |f(t)|dt$.

Proof.

(1) Let $\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_n = b\}$ be a partition of $[a,b]$. We have that

$$\begin{aligned} U(f + \lambda g, \mathcal{P}) &= \sum_{i=1}^{n} \sup_{x \in [x_{i-1}, x_i]} (f(x) + \lambda g(x))(x_i - x_{i-1}) \\ &\leq \sum_{i=1}^{n} \sup_{x \in [x_{i-1}, x_i]} f(x)(x_i - x_{i-1}) + \lambda \sum_{i=1}^{n} \sup_{x \in [x_{i-1}, x_i]} g(x)(x_i - x_{i-1}) \\ &= U(f,\mathcal{P}) + \lambda U(g,\mathcal{P}). \end{aligned}$$

Similarly, $L(f+\lambda g,\mathcal{P}) \geq L(f,\mathcal{P}) + \lambda L(g,\mathcal{P})$. We therefore have

$$L(f) + \lambda L(g) \leq L(f+\lambda g) \leq U(f+\lambda g) \leq U(f) + \lambda U(g).$$

Since $L(f) = U(f)$ and $L(g) = U(f)$, $L(f+\lambda g) = U(f+\lambda g) = U(f) + \lambda U(g)$.

(2) Let $\mathcal{P}$ be a partition as in the proof of (1). Then,

$$U(f,\mathcal{P}) = \sum_{i=1}^{n} \sup_{x\in[x_{i-1},x_i]} f(x)(x_i - x_{i-1}) \geq 0.$$

(3) Note that λ is integrable, as it is continuous. If $\mathcal{P}$ is a partition as described in the proof of (1), then

$$\begin{aligned} U(f,\mathcal{P}) &= \sum_{i=1}^{n} \sup_{x\in[x_{i-1},x_i]} \lambda(x_i - x_{i-1}) = \sum_{i=1}^{n} \lambda(x_i - x_{i-1}) \\ &= \lambda(b-a). \end{aligned}$$

(4) Let $U_{[x,y]}(f)$ denote $U(f)$ on the interval $[x,y]$. We claim that to find $U_{[a,b]}(f)$, we only need to take the infimum over partitions containing c. Suppose otherwise, that there was some larger lower bound of these partitions U'. Then there is some partition $\mathcal{P}$ with $U(f,\mathcal{P}) < U'$. Since $U(f,\mathcal{P}\cup\{c\}) \leq U(f,\mathcal{P})$, we have a contradiction. Similarly, $L_{[a,b]}(f)$ is the supremum over partitions containing c. Now, for each partition $\mathcal{P}$ with $c \in \mathcal{P}$, we have that $U_{[a,c]}(f,\mathcal{P}) + U_{[c,b]}(f,\mathcal{P}) = U_{[a,b]}(f,\mathcal{P})$, and similarly for L. Thus, $U_{[a,c]}(f) + U_{[c,b]}(f) \leq U_{[a,b]}(f) = L_{[a,b]}(f) \leq L_{[a,c]}(f) + L_{[c,b]}(f)$. However, we also have that $U_{[a,c]}(f) + U_{[c,b]}(f) \geq L_{[a,b]}(f) + L_{[c,b]})(f)$, meaning that $U_{[a,c]}(f) + U_{[c,b]}(f) = L_{[a,c]}(f) + L_{[c,b]}(f)$ (which must be equal to $U_{[a,b]}(f) = L_{[a,b]}(f)$). Now, suppose that $U_{[a,c]}(f) > L_{[a,c]}(f)$. Since $U_{[c,b]}(f) \geq L_{[c,b]}(f)$, we have that $U_{[a,b]}(f) = U_{[a,c]}(f) + U_{[c,b]}(f) > L_{[a,c]}(f) + L_{[c,b]}(f) = L_{[a,b]}(f)$, a contradiction. Thus, f is integrable on $[a,c]$. Similarly, it is integrable on $[b,c]$. We have already proved that $U_{[a,c]}(f) + U_{[c,b]}(f) = U_{[a,b]}(f)$, which is equivalent to $\int_a^c f(t)dt + \int_c^b f(t)dt = \int_a^b f(t)dt$.

(5) Notice that $-|f| \leq f \leq |f|$. Thus, $0 \leq f + |f|$ and $0 \leq |f| - f$. Thus, if we show that $|f|$ is integrable, then

$$\begin{aligned} 0 &\leq \int_a^b f(t) + |f(t)| dt \\ &= \int_a^b f(t)dt + \int_a^b |f(t)|dt, \end{aligned}$$

so $-\int_a^b |f(t)|dt \leq \int_a^b f(t)dt$. Similarly, $\int_a^b f(t)dt \leq \int_a^b |f(t)|dt$. Putting these together, $\left|\int_a^b f(t)dt\right| \leq \int_a^b |f(t)|dt$. Now, we show that $|f|$ is integrable. By assumption, f is integrable. Let $\varepsilon > 0$ and take a partition $\mathcal{P}$ so that $U(f,\mathcal{P}) - L(f,\mathcal{P}) < \varepsilon$. Write $\mathcal{P}$

in the form presented in the proof of (1). We have 3 cases. If $f \geq 0$ on $[x_{i-1}, x_i]$, then $f = |f|$ on that interval, and

$$\sup_{x \in [x_{i-1}, x_i]} f(x) - \inf_{x \in [x_{i-1}, x_i]} f(x) = \sup_{x \in [x_{i-1}, x_i]} |f(x)| - \inf_{x \in [x_{i-1}, x_i]} |f(x)|.$$

If $f(x) \leq 0$ on $[x_{i-1}, x_i]$, then $-f = |f|$ on that interval, and so

$$\begin{aligned}\sup_{x \in [x_{i-1}, x_i]} f(x) - \inf_{x \in [x_{i-1}, x_i]} f(x) &= \sup_{x \in [x_{i-1}, x_i]} -|f(x)| - \inf_{x \in [x_{i-1}, x_i]} -|f(x)| \\ &= -\inf_{x \in [x_{i-1}, x_i]} |f(x)| + \sup_{x \in [x_{i-1}, x_i]} |f(x)|.\end{aligned}$$

If the sign of f changes,

$$\max\left(\sup_{x \in [x_{i-1}, x_i]} f(x), -\inf_{x \in [x_{i-1}, x_i]} f(x)\right) = \sup_{x \in [x_{i-1}, x_i]} |f(x)|.$$

Thus,

$$\begin{aligned}\sup_{x \in [x_{i-1}, x_i]} |f(x)| - \inf_{x \in [x_{i-1}, x_i]} |f(x)| &\leq \max\left(\sup_{x \in [x_{i-1}, x_i]} f(x), -\inf_{x \in [x_{i-1}, x_i]} f(x)\right) \\ &\leq \sup_{x \in [x_{i-1}, x_i]} f(x) - \inf_{x \in [x_{i-1}, x_i]} f(x).\end{aligned}$$

In all three cases, we have

$$\sup_{x \in [x_{i-1}, x_i]} |f(x)| - \inf_{x \in [x_{i-1}, x_i]} |f(x)| \leq \sup_{x \in [x_{i-1}, x_i]} f(x) - \inf_{x \in [x_{i-1}, x_i]} f(x).$$

Then, $U(|f|, \mathcal{P}) - L(|f|, \mathcal{P}) \leq U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$.

$\blacksquare$

Definition 5.33. Suppose that f is integrable on $[a, b]$. We define

$$\int_b^a f(t) ft = -\int_a^b f(t) dt.$$

Theorem 5.34. *Suppose $f : [\alpha, \beta] \to \mathbb{R}$ is integrable, $a \in [\alpha, \beta]$, and set*

$$F : [\alpha, \beta] \to \mathbb{R}$$

$$x \mapsto \int_a^x f(t) dt.$$

Then F is continuous, and if f is continuous at $x_0 \in (\alpha, \beta)$, then F is differentiable there and $F'(x_0) = f$.

Proof. If $t, s \in [\alpha, \beta]$, then

$$F(t) = \int_a^s f(x)dx + \int_s^t f(x)dx = F(s) + \int_s^t f(x)dx.$$

If we show that $\lim_{t\to s} \int_s^t f(x)dx = 0$, we will conclude that $\lim_{t\to s} F(t) = F(s)$, so F is continuous (at s, and hence on $[\alpha, \beta]$). Now,

$$0 \le \left| \int_s^t f(x)dx \right| \le \sup_{x\in[s,t]} |f(x)| \cdot |t-s| \le \sup_{x\in[\alpha,\beta]} |f(x)| \cdot |t-s|.$$

Since $\lim_{t\to s} 0 = 0 = \lim_{t\to s} \left(\sup_{x\in[\alpha,\beta]} |f(x)| \right) \cdot |t-s|$, the Squeeze Theorem tells us

$$\lim_{t\to s} \left| \int_s^t f(x)dx \right| = 0,$$

so

$$\lim_{t\to s} \int_s^t f(x)dx = 0,$$

and $\lim_{t\to s} F(t) = F(s)$. Now, suppose that f is continuous at $x_i \in [\alpha, \beta]$, and $x \in [\alpha, \beta]$ is not x_0. We show that

$$\lim_{t\to x_0} \frac{F(t) - F(x_0)}{t - x_0} - f(x_0) = 0.$$

We have that this is equal to

$$\begin{aligned}
&\frac{1}{t-x_0} \left(\int_a^t f(x)dx - \int_a^{x_0} f(x)dx - f(x_0)(t-x_0) \right) \\
&= \frac{1}{t-x_0} \left(\int_{x_0}^t f(x)dx - f(x_0) \int_{x_0}^t 1dx \right) \\
&= \frac{1}{t-x_0} \int_{x_0}^t f(x) - f(x_0)dx.
\end{aligned}$$

Let $\varepsilon > 0$, choose $\delta > 0$ so that $(x_0 - \delta, x_0 + \delta) \subseteq [\alpha, \beta]$, and that $|f(x) - f(x_0)| < \varepsilon$ if $|x - x_0| < \delta$. Then, for $t \in (x_0 - \delta, x_0 + \delta)$, we have

$$\begin{aligned}
\left| \frac{F(t) - F(x_0)}{t - x_0} - f(x_0) \right| &= \frac{1}{|t-x_0|} \left| \int_{x_0}^t f(x) - f(x_0)dx \right| \\
&\le \frac{1}{|t-x_0|} \sup_{x\in[x_0,t]} |f(x) - f(x_0)| \cdot |t - x_0| \\
&< \varepsilon.
\end{aligned}$$

Thus, $F'(x_0) = \lim_{t\to x_0} \frac{F(t) - F(x_0)}{t - x_0} = f(x_0)$. ■

Theorem 5.35 (Fundamental Theorem of Calculus). *Suppose $f : [a, b] \to \mathbb{R}$ is integrable and that there is some $F : [a, b] \to \mathbb{R}$ continuous and differentiable on (a, b) with $F' = f$ on (a, b). Then,*

$$\int_a^b f(t)ft = F(b) - F(a).$$

Proof. Let $\varepsilon > 0$, and choose a partition $\mathcal{P}$ so that $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$. Now, write $\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_n = b\}$. Then, by the Mean Value Theorem, $F(x_i) - F(x_{i-1}) = F'(t_i)(x_i - x_{i-1}) = f(t_i)(x_i - x_{i-1})$ for some $t_i \in (x_{i-1}, x_i)$. However,

$$L(f, \mathcal{P}) \leq \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1}) = \sum_{i=1}^{n} F(x_i) - F(x_{i-1}) = F(b) - F(a) \leq U(f, \mathcal{P}).$$

Now, $|L(f, \mathcal{P}) - L(f)| < \varepsilon$, and also $|L(f, \mathcal{P}) - (F(b) - F(a))| < \varepsilon$. Thus, $|L(f) - (F(b) - F(a))| < 2\varepsilon$. Since ε is arbitrary, $\int_a^b f(t)dt = L(f) = F(b) - F(a)$. ■

6 Sequences of Functions

6.1 Pointwise Convergence

Definition 6.1. Suppose that $(f_n)_n$ is a sequence of functions between metric spaces $X \to Y$, We say $(f_n)_n$ converges pointwise to a function $f : X \to Y$ if for every $x \in X$, the sequence $(f_n(x))_n$ converges to $f(x)$.

Example 6.2. Let $f_n : [0, 1] \to [0, 1]$ be defined by $t \mapsto t^n$. Now if $t \in [0, 1)$, then $\lim_{n\to\infty} f_n(t) = \lim_{n\to\infty} t^n = 0$. If $t = 1$, then $\lim_{n\to\infty} f_n(t) = \lim_{n\to\infty} t^n = 1$. Thus, $(f_n)_n$ converges pointwise to $f : [0, 1] \to \mathbb{R}$ defined by

$$f(t) = \begin{cases} 0 & t \in [0, 1) \\ 1 & t = 1 \end{cases}.$$

This shows that both continuity and differentiability are not preserved under pointwise convergence.

Example 6.3. Let $(q_n)_n$ be a sequence of rational numbers so that every element of $\mathbb{Q} \cap [0, 1]$ occurs at least once. Set $g_n : [0, 1] \to \mathbb{R}$ be defined by

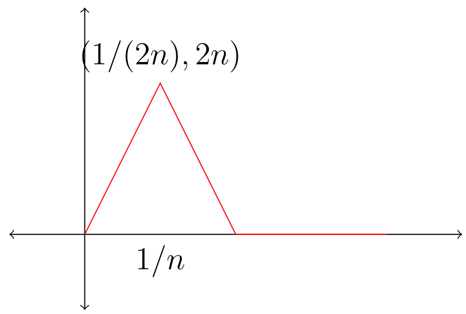
$$t \mapsto \begin{cases} 1 & t = q_k \text{ for some } k < n \\ 0 & \text{else} \end{cases}.$$

Then $(g_n)_n$ converges pointwise to the function $g : [0, 1] \to \mathbb{R}$ defined by

$$t \mapsto \begin{cases} 1 & t \in \mathbb{Q} \\ 0 & t \notin \mathbb{Q} \end{cases}.$$

Then, $\int_0^1 g_n(t)dt = 0$, while g is not integrable, meaning that integrability is not preserved under pointwise convergence.

Example 6.4. Let $h_n : [0,1] \to \mathbb{R}$ be defined as in the following picture.



Then $\int_0^1 h_n(t)dt = 1$. However, $\lim_{n\to\infty} h_n(t) = 0$ for every $t \in [0,1]$. Thus, even if integrability is conserved, the limit of the integrals is not equal to the integral of the limit.

6.2 Uniform Convergence

Definition 6.5. Suppose that S is a set, X is a metric space, and $(f_n)_n$ is a sequence of functions from $S \to X$. We say that $(f_n)_n$ converges uniformly to a limit $f : S \to X$ if for every $\varepsilon > 0$, there is some $N \in \mathbb{N}$ so that for all $s \in S$ and $n > N$, $d_X(f_n(s), f(s)) < \varepsilon$.

Remark 6.6. None of our previous examples of pointwise convergence converge uniformly.

Theorem 6.7. *Suppose that $(f_n)_n$ is a sequence of continuous functions from metric spaces $X \to Y$ converging uniformly to $f : X \to Y$. Then f is continuous.*

Proof. Let $x \in X$, and let $\varepsilon > 0$. Choose n so that for every $t \in X$, we have $d_Y(f_n(t), f(t)) < \varepsilon/3$. Let $\delta > 0$ so that if $d_X(x,t) < \delta$, then $d_Y(f_n(x), f_n(t)) < \varepsilon/3$. Now, if $d_X(x,t) < \delta$, then

$$d_Y(f(x), f(t)) \leq d_Y(f(x), f_n(x)) + d_Y(f_n(x), f_n(t)) + d_Y(f_n(t), f_n(x)) < \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon.$$

■

Proposition 6.8. *Uniform convergence implies pointwise convergence to the same limit.*

Proof. Suppose $(f_n)_n$ is a sequence of functions from a set to a metric space $S \to X$ which converges uniformly to $f : S \to X$. Let $\varepsilon > 0$ and $x \in S$. Since $(f_n)_n$ converges uniformly, there is some $N \in \mathbb{N}$ so that for every $t \in S$ and $n > N$, $d_X(f_n(t), f(t)) < \varepsilon$. In particular, $d_X(f_n(x), f(x)) < \varepsilon$. Hence, $\lim_{n\to\infty} f_n(x) = f(x)$. ■

Definition 6.9. Suppose $(f_n)_n$ is a sequence of functions from a set S to a metric space X. The sequence is uniformly Cauchy if for every $\varepsilon > 0$, there is N so that for every $n, m > N$ and any $s \in S$, we have $d_X(f_n(s), f_m(s)) < \varepsilon$.

Theorem 6.10. *Suppose $(f_n)_n$ is a sequence of functions from a set S to a metric space X. If $(f_n)_n$ converges uniformly, it is uniformly Cauchy. Moreover, if X is complete, then the reverse implication is true.*

Proof. Let $\varepsilon > 0$, and suppose that $(f_n)_n$ converges uniformly to f. Choose N so that if $n > N$ and $s \in S$, then $d_X(f_n(s), f(x)) < \varepsilon/2$. Then, if $m > N$, $d_X(f_n(s), f_m(s)) \leq d_X(f_n(s), f(s)) + d_X(f(s), f_m(s)) < \varepsilon/2 + \varepsilon/2 = \varepsilon$. Now suppose that X is complete and that $(f_n)_n$ is uniformly Cauchy. Note that $(f_n(s))_n$ is Cauchy for every $s \in S$. Since X is complete, $\lim\limits_{n\to\infty} f_n(s)$ exists. Then we can define $f : S \to X$ by $s \mapsto \lim\limits_{n\to\infty} f_n(s)$. We claim that $(f_n)_n$ converges uniformly to f. Let $\varepsilon > 0$ Choose N so that if $n, m > N$, $d_X(f_n(s), f_m(s)) < \varepsilon/2$. Fix $s \in S$, and let $n_s > N$ be so that $d_X(f_{n_s}(s), f(s)) < \varepsilon/2$. Now if $m > N$ and $s \in S$, $d_X(f_m(s), f(s)) \leq d_X(f_m(s), f_{n_s}(s)) + d_X(f_{n_s}(s), f(s)) < \varepsilon/2 + \varepsilon/2 = \varepsilon$. ■

Definition 6.11. Suppose that S is a set and $f : S \to \mathbb{R}$. We define the uniform norm of f to be $||f||_\infty = \sup\{|f(s)| : s \in S\}$ if the supremum exists. If the supremum does not exist, then $||f||_\infty = \infty$. When emphasis is needed, we will write $||f||_{\infty,S}$.

Remark 6.12. $||\cdot||_\infty$ almost gives us a metric on functions $S \to \mathbb{R}$ by setting $d_\infty(f, g) = ||f - g||_\infty$. However, it can take on infinite values.

Proposition 6.13. *the sequence $(f_n)_n$ of functions $S \to \mathbb{R}$ converges uniformly to $f : S \to \mathbb{R}$ if and only if $\lim\limits_{n\to\infty} ||f_n - f||_\infty = 0$.*

Proof. Suppose that $(f_n)_n$ converges uniformly to f. Choose $\varepsilon > 0$, and choose $N \in \mathbb{N}$ so that for $n > N$ and all $s \in S$, $|f_n(s) - f(s)| < \varepsilon$. Then $||f_n - f||_\infty = \sup\{|f_n(s) - f(s)| : s \in S\} \leq \varepsilon$. Thus, $\lim\limits_{n\to\infty} ||f_n - f||_\infty = 0$. Conversely, suppose that $\lim\limits_{n\to\infty} ||f_n - f||_\infty = 0$. Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ so that for $n > N$, $||f_n - f||_\infty < \varepsilon$. Then $\sup\{|f_n(s) - f(s)| : s \in S\} < \varepsilon$, so that for all $s \in S$ and $n > N$, $|f_n(s) - f(s)| < \varepsilon$. Thus, $(f_n)_n$ converges uniformly to f. ■

Theorem 6.14. *Suppose $(f_n)_n$ is a sequence of integrable functions on $[a, b]$ which converge uniformly to a limit f. Then f is integrable, and $\lim\limits_{n\to\infty} \int_a^b f_n(t)dt = \int_a^b f(t)dt$.*

Proof. We first show that f is integrable. Let $\varepsilon > 0$. Choose $n \in \mathbb{N}$ so that $||f_n - f||_\infty < \varepsilon/(2(b - a))$. Then $f_n - \varepsilon/(2(b - a)) < f < f_n + \varepsilon/(2(b - a))$. Thus,

$$U(f) \leq U(f_n + \varepsilon/(2(b - a))) = \int_a^b f_n(t)dt + \frac{\varepsilon(b - a)}{2(b - a)} = \frac{\varepsilon}{2},$$

and

$$L(f) \geq L(f_n - \varepsilon/(2(b - a))) = \int_a^b f_n(t)dt - \frac{\varepsilon(b - a)}{2(b - a)} = \frac{\varepsilon}{2},$$

so $U(f) - L(f) \leq \frac{2\varepsilon}{2} = \varepsilon$. Thus, for any $\varepsilon > 0$, we have $U(f) - L(f) < \varepsilon$, so f is integrable. Now, for any $n \in \mathbb{N}$, we have

$$0 \leq \left| \int_a^b f(t)ft - \int_a^b f_n(t)dt \right| \leq \int_a^b |f(t) - f_n(t)|dt.$$

Let $\varepsilon > 0$. If N is large enough so that $||f_n - f||_\infty < \frac{\varepsilon}{b-a}$ for every $n > N$, we have $\int_a^b |f(t) - f_n(t)|dt \leq \int_a^b \frac{\varepsilon}{b-a} dt = \varepsilon$. Hence, $\lim\limits_{n\to\infty} \int_a^b f_n(t)dt = \int_a^b f(t)dt$. ■

Theorem 6.15 (Stone' Theorem)**.** *Suppose $f : [a,b] \to \mathbb{R}$ is continuous. Then there is a sequence of polynomials $(p_n)_n$ converging uniformly to f on $[a,b]$.*

Proof. We first prove that it suffices to prove this with $[a,b] = [0,1]$. If $f : [a,b] \to \mathbb{R}$ is continuous, then so is $g : [0,1] \to \mathbb{R}$ defined by $t \mapsto f(a+t(b-a))$, and if $(p_n)_n$ is a sequence of polynomials converging uniformly to g on $[0,1]$, then $q_n : x \mapsto p_n((x-a)/(b-a))$ is a sequence of polynomials converging uniformly to f on $[a,b]$.

Now, note that we can assume that $f(0) = 0 = f(1)$, as we may subtract $f(0) + tf(1)$ to obtain such a function, and add the same linear term back to our sequence of polynomials.

Finally, we extend f to $\mathbb{R}$ by setting $f(x) = 0$ if $x \notin [0,1]$.

For each n, define $\mathcal{Q}_n(t) = c_n(1-t^2)^n$, where $c_n = \left(\int_{-1}^1 (1-t^2)^n dt\right)^{-1}$. Then $\int_{-1}^1 \mathcal{Q}_n(t)dt = 1$. Set $\mathcal{P}_n(x) = \int_{-1}^1 f(x+t)\mathcal{Q}_n(t)dt = \int_{-x}^{1-x} f(x+t)\mathcal{Q}_n(t)dt$, where the second equality is true because f is 0 when it is not on $[0,1]$. This is equal to $\int_0^1 f(t)\mathcal{Q}_n(t-x)dt$, which is equal to

$$\int_0^1 f(t) \sum_{k=0}^{2n} a_k(t)x^k dt = \sum_{k=0}^{2n} \left(\int_0^1 f(t)a_k(t)dt\right) x^k,$$

where a_k is chosen as necessary. Hence $\mathcal{P}_n$ is a polynomial. We claim $(\mathcal{P}_n)_n$ converges uniformly to f.

Let $\varepsilon > 0$. Since f is uniformly continuous on $[-10,10]$, there is some $1 > \delta > 0$ so that if $x, y \in [-10,10]$ with $|x-y| < \delta$, then $|f(x) - f(y)| < \varepsilon/2$. Let $x \in [0,1]$. Then,

$$\begin{aligned}
|\mathcal{P}_n(x) - f(c)| &= \left|\int_{-1}^1 f(x+t)\mathcal{Q}_n(t)dt - f(x)\int_{-1}^1 \mathcal{Q}_n(t)dt\right| \\
&\leq \int_{-1}^1 |f(x+t) - f(x)|\mathcal{Q}_n(t)dt. \\
&= \int_{-1}^{-\delta} |f(x+t) - f(x)|\mathcal{Q}_n(t)dt + \int_\delta^1 |f(x+t) - f(x)|\mathcal{Q}_n(t)dt \\
&+ \int_{-\delta}^{\delta} |f(x+t) - f(x)|\mathcal{Q}_n(t)dt.
\end{aligned}$$

Now, $\int_{-\delta}^{\delta} |f(x+t) - f(x)|\mathcal{Q}_n(t)dt < \int_{-\delta}^{\delta} \varepsilon/2\mathcal{Q}_n(t)dt \leq \int_{-1}^1 \varepsilon/2\mathcal{Q}_n(t)dt = \varepsilon/2$. Now,

$$\begin{aligned}
&\int_{-1}^{-\delta} |f(x+t) - f(x)|\mathcal{Q}_n(t)dt + \int_\delta^1 |f(x+t) - f(x)|\mathcal{Q}_n(t)dt \\
&\leq \int_{-1}^{-\delta} 2||f||_\infty \mathcal{Q}_n(t)dt + \int_\delta^1 2||f||_\infty \mathcal{Q}_n(t)dt.
\end{aligned}$$

Thus, we need to control $\int_\delta^1 \mathcal{Q}_n(t)dt$ and $\int_{-1}^{-\delta} \mathcal{Q}_n(t)dt$. We have that $\mathcal{Q}(t) = c_n(1-t^2)^n$. We have that

$$c_n^{-1} = \int_{-1}^1 (1-t^2)^n dt \geq \int_{-1/\sqrt{n}}^{1/\sqrt{n}} (1-t^2)^n dt,$$

which we claim is at least $\int_{-1/\sqrt{n}}^{1/\sqrt{n}} 1-nt^2 dt$. Note that it suffices to show that $(1-s)^n \geq 1-ns$ on $[0,1/n]$, which is clear. Now, this bound is $4/3\sqrt{n} > 1/\sqrt{n}$. If $|t| \geq \delta$, then $\mathcal{Q}_n(t) = c_n(1-t^2)^n < \sqrt{n}(1-\delta^2)^n$, meaning that $\int_\delta^1 \mathcal{Q}_n(t)dt < \int_1^\delta \sqrt{n}(1-\delta^2)^n dt = \sqrt{n}(1-\delta^2)^n(1-\delta)$, which goes to 0 as we make n large. ∎